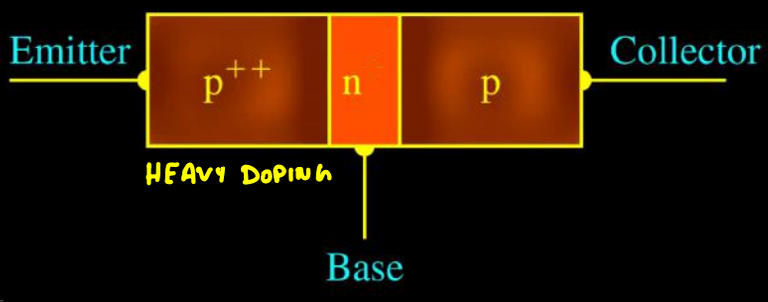
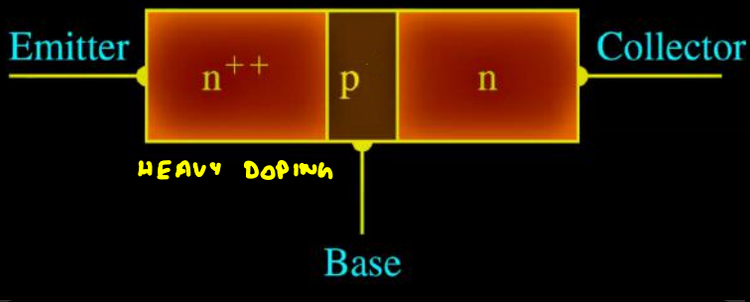
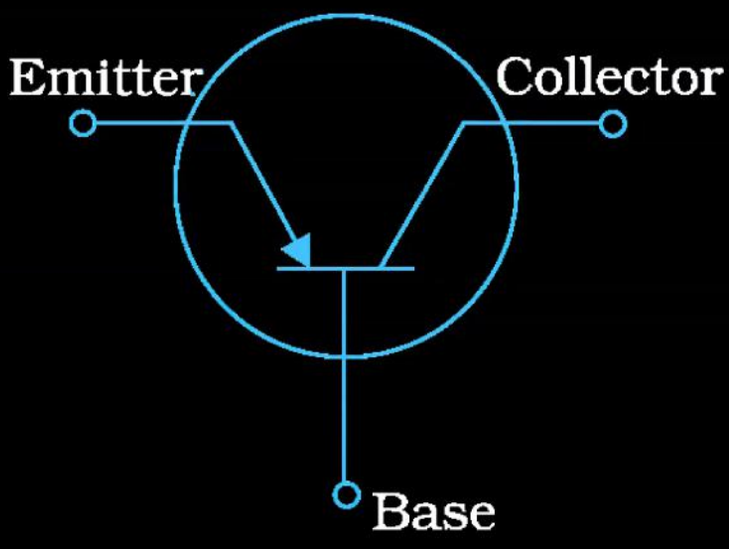
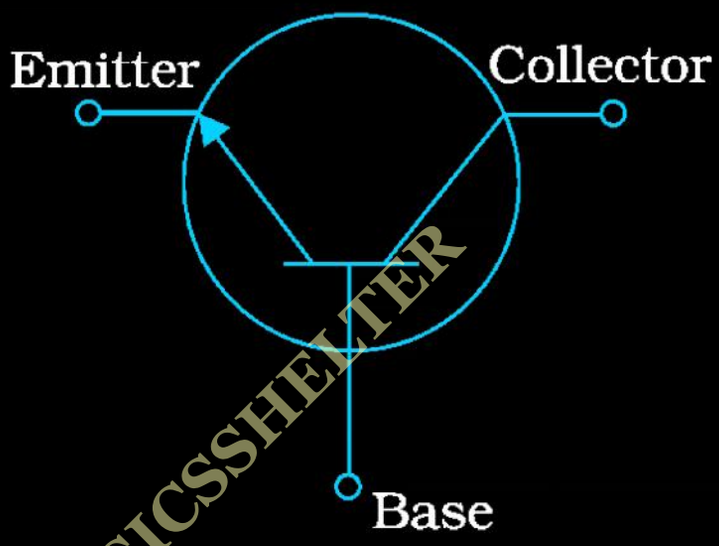
Transistor Structure And Symbols:









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A transistor has 3 doped regions forming 2 p-n junctions as seen in the figure above. There are two types of transistors:

npn transistor: Here two segments of n-type (emitter and collector) are separated by a p type crystal (base).

pnp transistor: Here two segments of p type semi-conductors (termed as emitter and collector) are separated by an n type crystal (known as base). The transistors are made by doping same crystal with different types of dopants in different places.

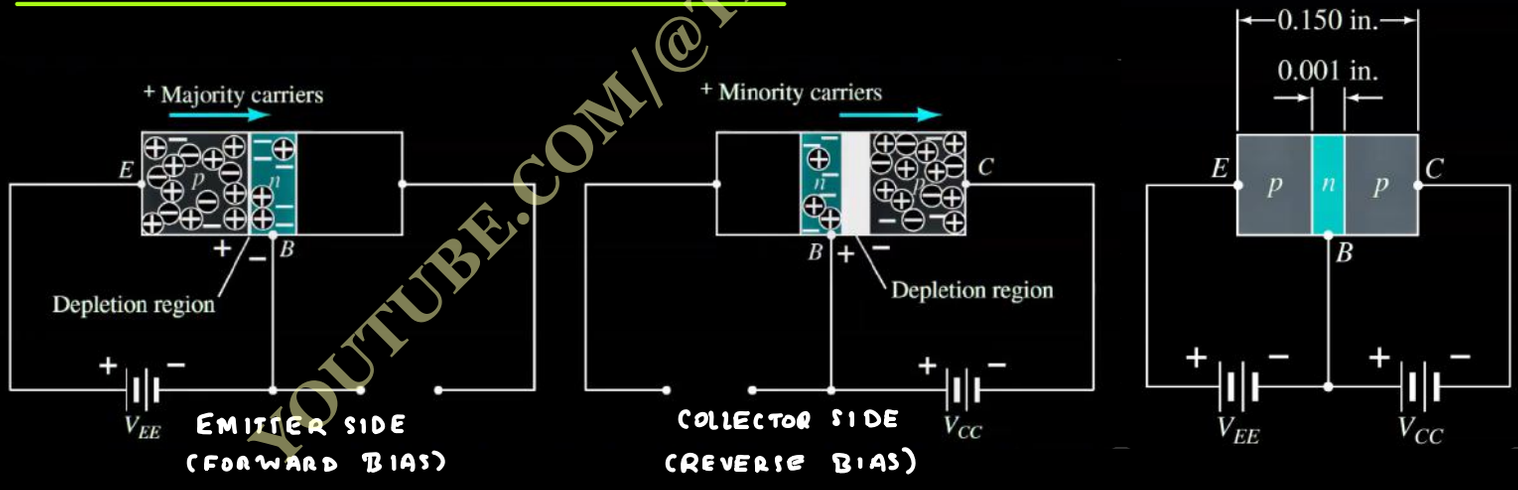
Transistor Symbols: The transistor symbols for the two types of transistors are shown in the figure above. The emitter is always marked with an arrow to denote the direction of conventional current. In a typical transistor application, the majority carriers flow from the emitter towards the base hence the arrow heads (student should verify it in the symbols).

EMITTER: This segment is on one side of the transistor and it is of moderate size and heavily doped. Its function is to supply a large number of majority carriers for current flow through the transistor.

BASE: This is the central segment of the transistor and it is very thin and lightly doped (as per NCERT, though some advanced texts like Donald Neamen disagree). Most emitted majority carriers cross the base straight in to collector due to its thin size and light doping. (In a typical transistor application).

COLLECTOR: This segment collects the major portion of the majority carriers supplied by the emitter. It is moderately doped (as per NCERT but Donald Neamen Disagrees), and larger in size as compared to emitter.

TRANSISTOR OPERATION (COMMON BASE):

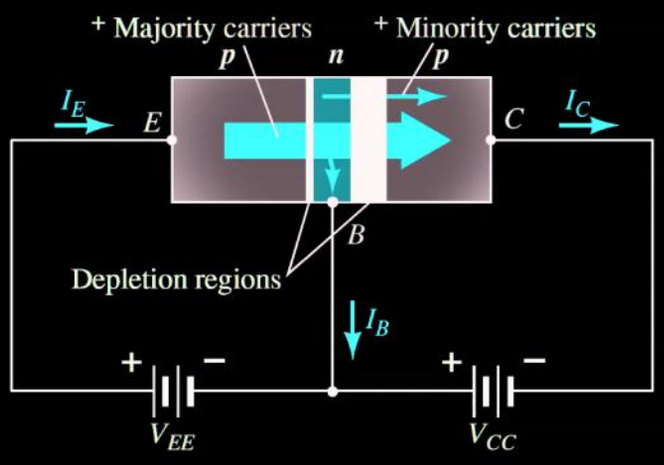


As described earlier, a transistor consists of two p-n junctions, one involving the emitter and the other involving the collector. In a typical transistor operation, emitter side p-n junction is forward biased while the collector side p-n junction is reverse biased.

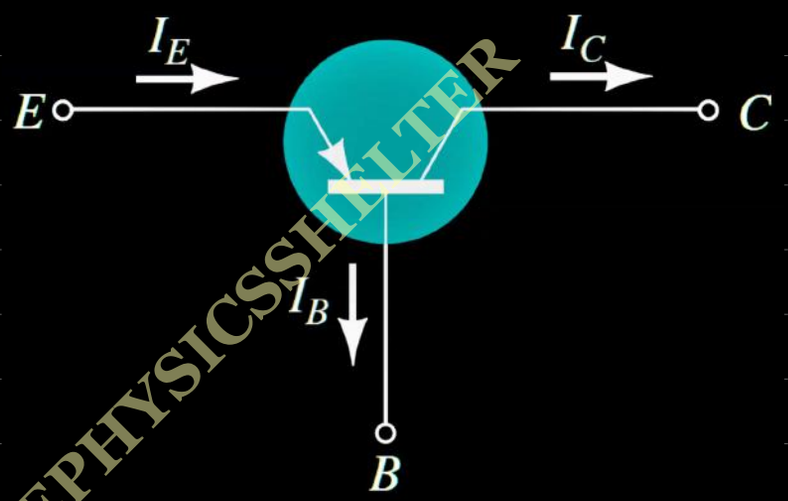
We will try to understand transistor for a pnp transistor (for npn functioning is identical with roles of electron and holes swapped). To better understand transistor action, we will consider biasing one junction at a time. The first figure above shows forward biasing of base-emitter

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junction. Here it behaves almost like a forward biased diode. The depletion layer is very thin, and large current will flow due to movement of majority carriers. In the second figure, the base collector junction is reverse biased. Recall that a reverse biased junction is favorable for minority carrier current, (holes are minority carriers in n type crystals). Thus there just like a typical reverse biased p-n junction, there will be a wide depletion region and some small current would flow due to movement of minority carriers.



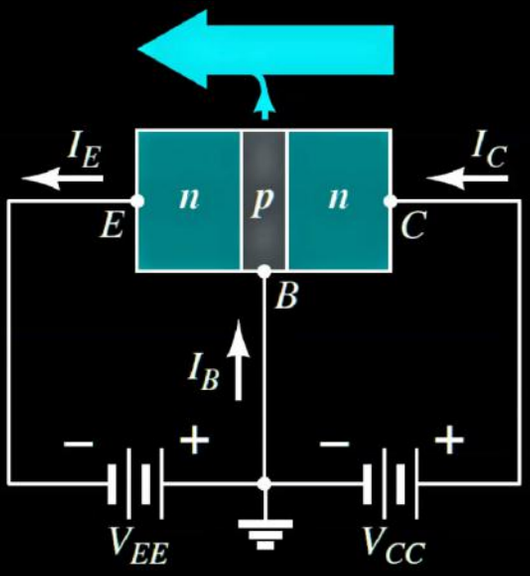
Majority and minority carrier flow of a pnp transistor.

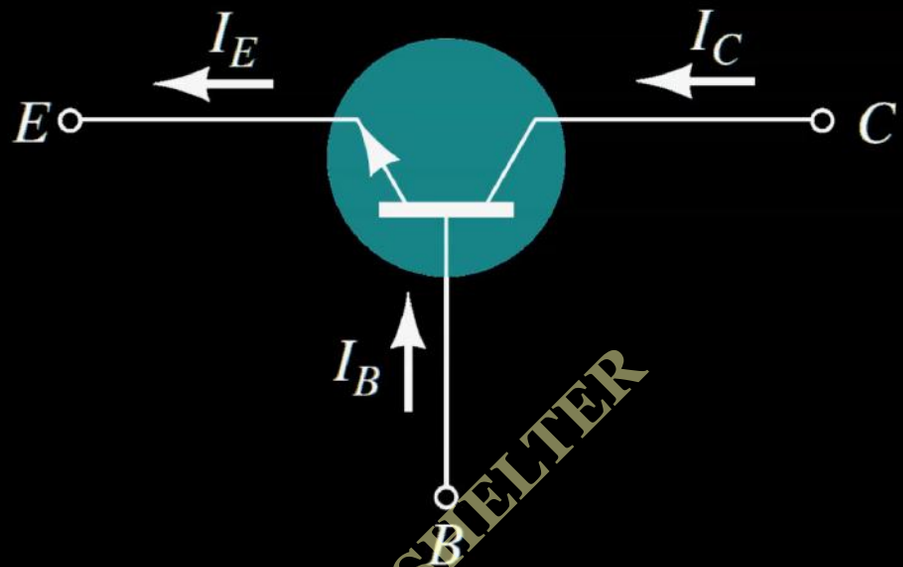


Now when both the biasing batteries are simultaneously applied, the majority and minority carrier flows are as shown in the figure above. The emitter emits a large number of holes in to the base where they are minority carriers and most of these holes are readily swept in to the collector (recall that a reverse bias is favorable for minority carrier current). This can also be understood directly. The base is very thin so most holes will easily pass in to collector where they are being "welcomed" by the collector side battery. Very few of the holes passing in to the base may combine with electrons there, giving rise to a small amount of base current. In addition to the majority carrier current arising due to swamping of the base by the majority carriers coming from the emitter, there is also some minority carrier current just like in a usual reverse biased diode. [Some holes in valence band of n-crystal will pass on to the p side (collector)]. The net current is the superposition of the majority carrier current and minority carrier currents.

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Typically, the base current is in micro-amperes and the collector and emitter currents are in mA. Applying junction law for the transistor we can write $I_E = I_B + I_C$

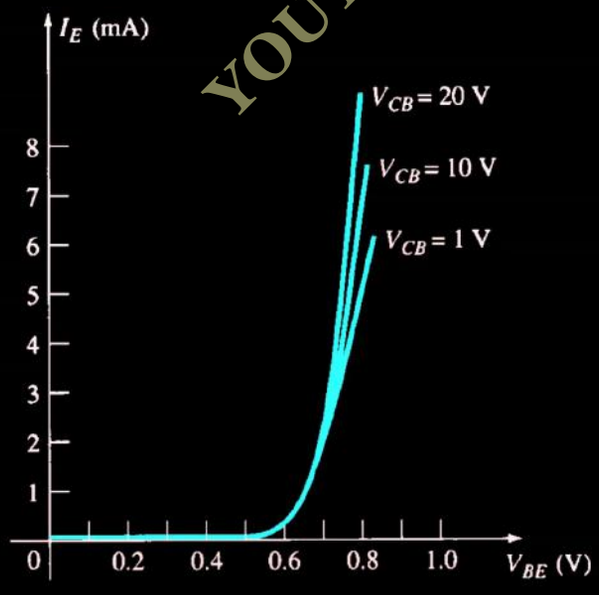




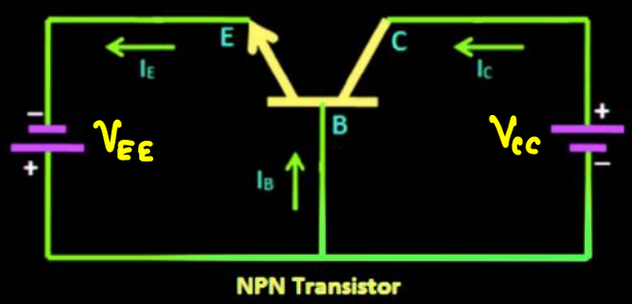
For an n-p-n transistors, both majority and minority currents will be just opposite to that of a p-n-p transistor as can be seen above.

TRANSISTOR CIRCUIT CONNECTIONS: A transistor is a three terminal device with two battery connections. Thus, one of the terminals has to be common between two biasing batteries. Depending on the common terminal, a transistor connection may be called:

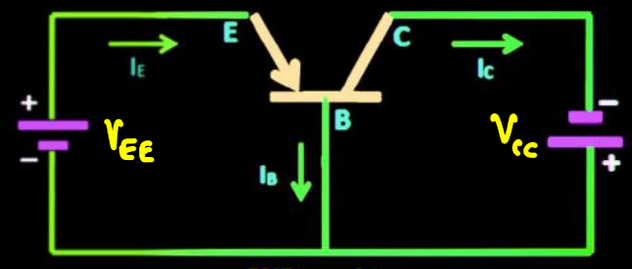
COMMON BASE:



Input or driving point characteristics for a common-base silicon transistor amplifier.



NPN Transistor

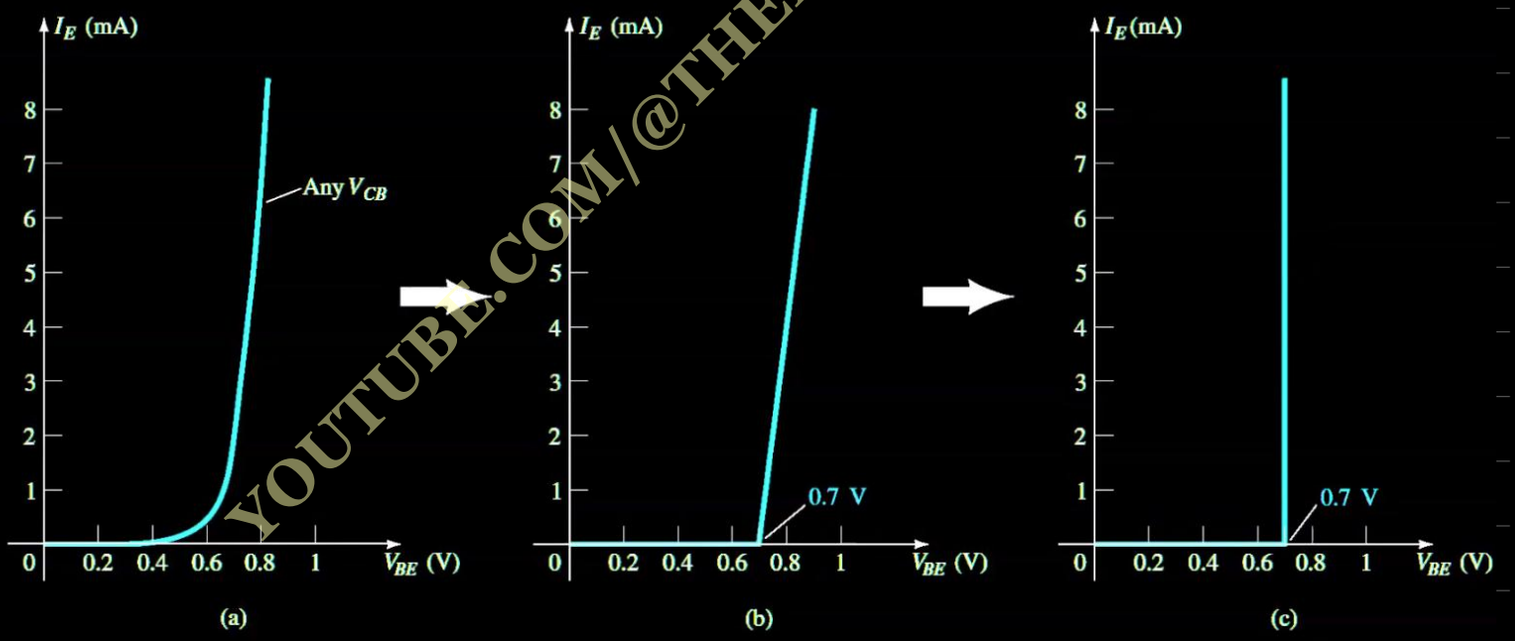


PNP transistor

Common base configuration

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As the name suggests, in common base configuration, the base is the common terminal between the two biasing batteries. (So far in the previous section we were only drawing common base circuits without mentioning the name). The first figure above shows the input characteristics of a common base configuration which represents emitter current against various voltage drops across base emitter junction (V_{BE}) for various values of base collector voltages (V_{BC}). Notice that the characteristic curves is almost like that of a forward biased diode. The base collector voltage has a very small effect on emitter current though with increase in collector side voltage (reverse) emitter current increases slightly. As usual we observe a knee voltage of about 0.7 Volts. Common base characteristics are not very relevant to JEE so the output characteristics for this configuration are not being discussed. (Output characteristics are a plot of collector current at various base-collector voltages).



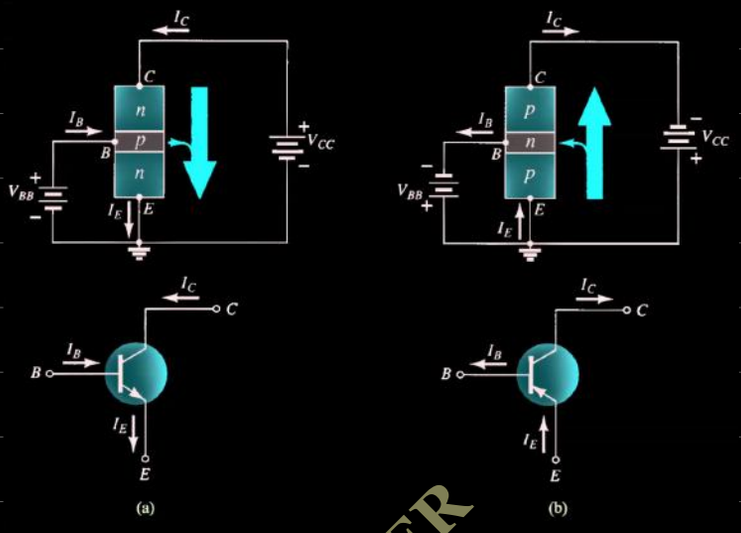
Developing the equivalent model to be employed for the base-to-emitter region of an amplifier in the dc mode.

Since base emitter junction is almost like a forward biased diode, just like a silicon diode, when base emitter junction is properly forward biased, we can assume base emitter voltage to be 0.7 Volts. (This is true for any of the three connections: Common base, common emitter and common collector)

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Common Emitter Configuration:

As the name suggests, here emitter is made the common terminal between the two biasing batteries. The common emitter configuration is most commonly used configuration and discussed in most detail in content of JEE. We will be learning about its detailed characteristics in a later section.

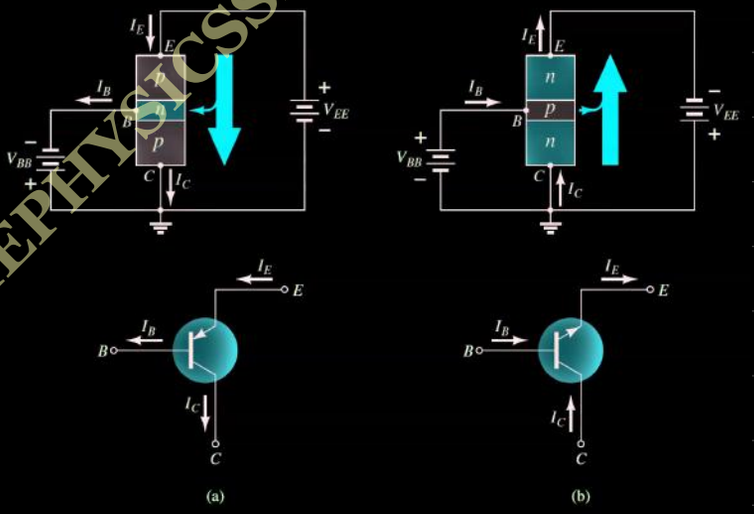


Notation and symbols used with the common-emitter configuration: (a) npn transistor; (b) pnp transistor.

Common Collector Configuration:

Here, as the name suggests, the collector is made the common terminal of the two biasing batteries.

The connections and current flows for common collector configuration are illustrated in adjacent figure. Common collector characteristics are not discussed in the NCERT however students are expected to solve numerical involving common collector also.

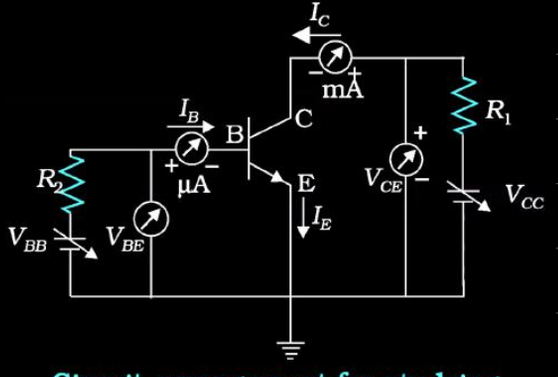


Notation and symbols used with the common-collector configuration: (a) pnp transistor; (b) npn transistor.

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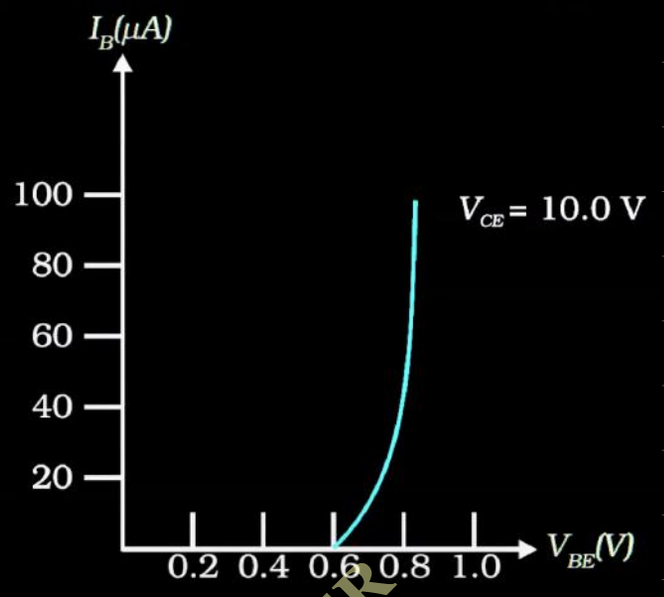
Common Emitter Input Characteristics:

A graph of base current vs base emitter voltage at some constant value of emitter collector voltage is called input characteristics. As discussed earlier, most of the majority carriers emitted by the emitter pass in to the collector and a few combine with minority carriers in the base to form the base

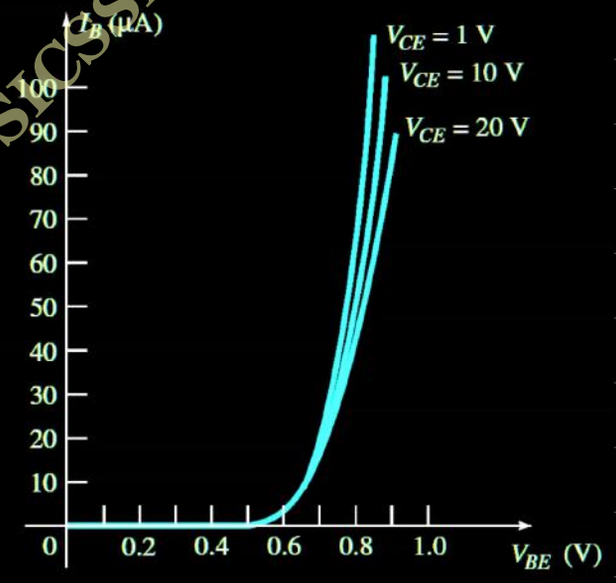


Circuit arrangement for studying the input and output characteristics of n-p-n transistor in CE configuration.

current. When the base emitter junction is properly forward biased, the emitter current behaves like that of a forward biased diode, however since most electrons are passing in to the collector, the base current is still in microamperes. With increase in collector base voltage i.e "Welcoming Voltage towards base", the base current decreases slightly though with sufficiently high V_{CE}, the base current maintains nearly a constant ratio with collector current. i.e $I_C/I_B \approx$ Constt. Just like a diode one can clearly see that until base emitter voltage is about 0.6 - 0.7 volts, the base current is practically zero (and so also the collector current). When the base and collector current are nearly zero due to insufficient emitter bias (< 0.6 Volts), the transistor is said to be cut off.). Infact if base emitter voltage is very low, base current can even go negative due to dominance of minority carrier current. This is possible in the cut off region. For example in pnp transistor holes flow downwards from the base when emitter is properly biased because holes are swamping the base, however if there is no biasing battery in the emitter, the base collector junction is just like a normal reverse biased diode with holes from base (minority) moving towards collector.



INPUT CHARACTERISTICS (NCERT)



INPUT CHARACTERISTICS (Boylestad)

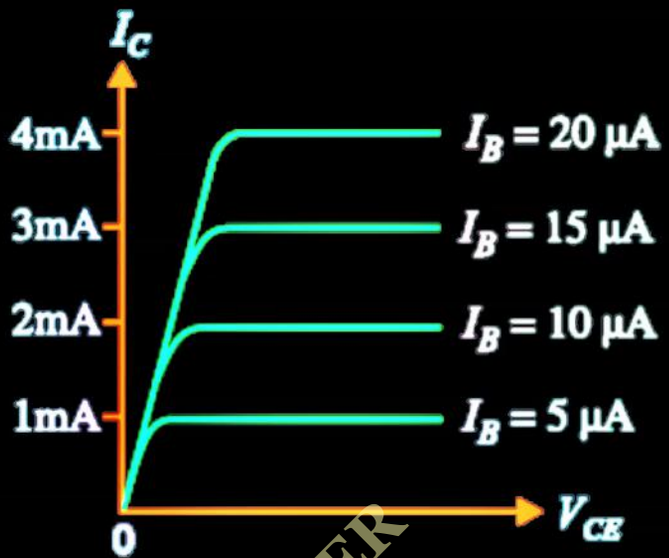
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Common Emitter Output Characteristics:

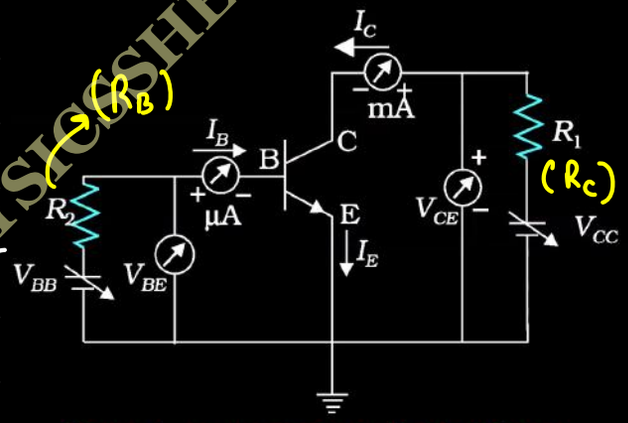
A graph of collector current vs collector-Emitter voltage V_{CE} at constant value of base current is called output characteristics. When the base emitter junction is properly forward biased and collector emitter voltage is large enough to "welcome" almost all the emitted electrons, the transistor is said to be in an active state. In the active state, the ratio of collector current to the base current is nearly constant. This ratio is called β of the transistor. For example in the above characteristic graph,

$$\beta = \left(\frac{I_C}{I_B}\right)_{active} = \frac{4mA}{20\mu A} = \frac{2mA}{10\mu A} = \frac{3mA}{15\mu A} = 200.$$

We can see that value of β comes out to be same from different curves drawn at different values of I_B. Thus $\boxed{\beta = \frac{I_C}{I_B}}$. Typically for most transistors β will lie somewhere between 20 and 200.

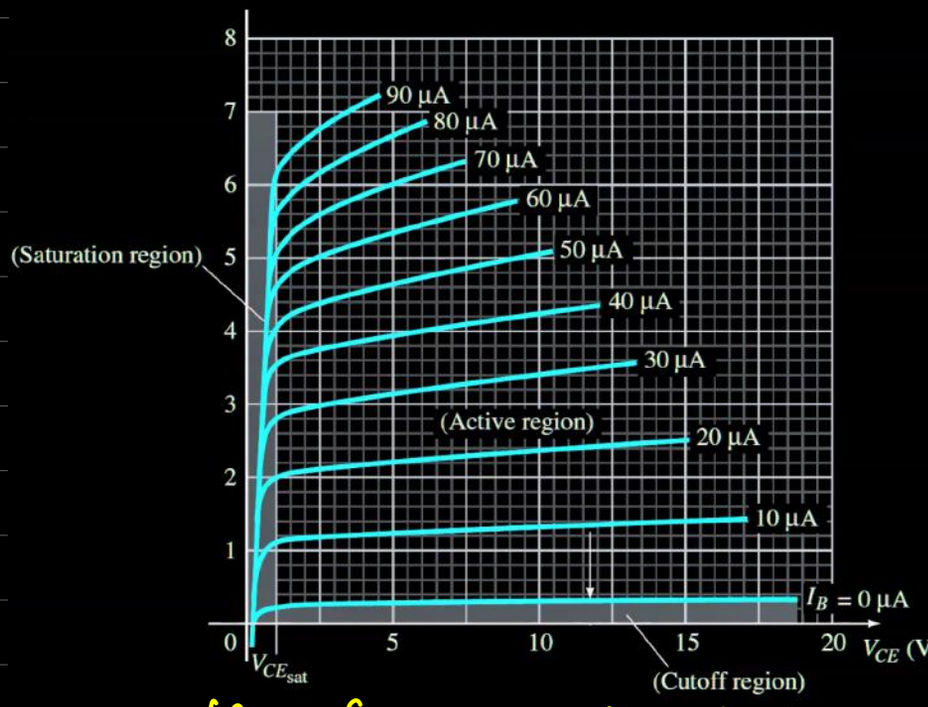


Output Characteristics (Ideal).



Circuit arrangement for studying the input and output characteristics of n-p-n transistor in CE configuration.

Saturation: From the transistor circuit we can see that, $V_{CE} = V_{CC} - I_C R_C$. Now for a given value of V_{CC} if emitter current becomes very large, I_C also becomes very large and then V_{CE} becomes nearly zero. When this happens, the transistor is said to be saturated. This is represented by left region in the graph.



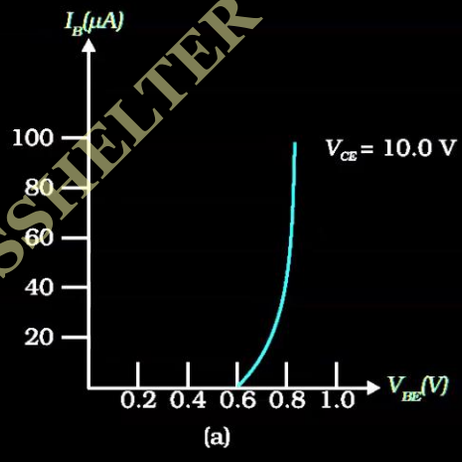
(Output Characteristics (Boylestad))

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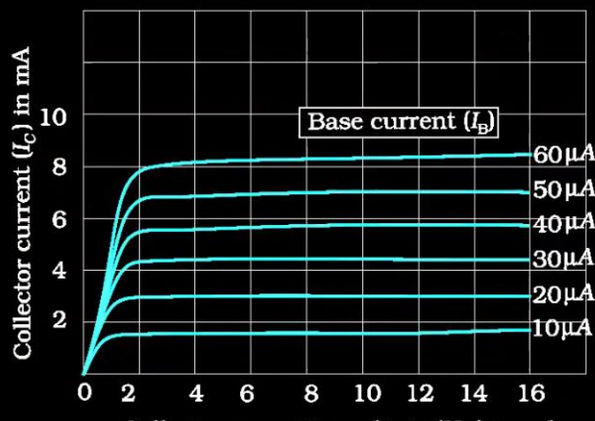
CUT OFF: When base emitter junction is not properly biased, the emitter is not able to emit properly. When this happens, base current is practically zero (or even negative due to reverse bias minority carrier current of the base collector junction). When this happens, the transistor is said to be in cut-off region. (See the bottom region of characteristics curve (Boylestad)).

AC PARAMETERS OF TRASISTOR CE CHARACTERISTICS:

In several applications, transistors are subjected to time varying voltages and currents fluctuating in the vicinity of some mean voltage (operating point). In such cases it is often useful to define dynamic or AC parameters of the transistors. (This is mostly done in the active regions where characteristic curves can be modeled as nearly straight lines). Now we will define some A.C. Parameters or dynamic parameters.



(a)



(b)

(a) Typical input characteristics, (b) Typical output characteristics.

INPUT RESISTANCE (R_i): The ratio of change in base-emitter voltage to change in base current at constant value of V_{CE} is called input resistance. This is dynamic (a.c.) resistance and as can be seen from input characteristic curve, its value varies with operating current in the transistor. Mathematically:

$$R_i = \left(\frac{\Delta V_{BE}}{\Delta I_B}\right)_{V_{CE}}$$

← See input characteristics Graph.

In typical transistor applications, input resistance can be anywhere between few hundred to few thousand ohms.

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Output Resistance (R_o): The ratio of change in collector emitter voltage (ΔV_{CE}) to the change in collector current (ΔI_C) at constant value of base current. Mathematically, $\boxed{R_o = \left(\frac{\Delta V_{CE}}{\Delta I_C}\right)_{I_B}}$

At very low values of V_{CE} i.e "Welcoming" voltage, we say that initially, I_C is linearly proportional to the V_{CE}. This is because base collector junction is not sufficiently biased and the electrons are not sufficiently motivated to cross the long and high resistance collector region. (The transistor is in saturation mode). However when V_{CE} is sufficiently large, I_C becomes nearly constant. The reciprocal of slope of nearly flat part of output characteristics gives the value of R_o. Thus output resistance is mainly controlled by the biasing battery on the collector side i.e V_{cc}. Thus in active state, magnitude of output resistance is quite high (of the order of 100 kΩ) and in saturation region it is relatively much smaller. (This is somewhat like characteristics of reverse biased diode where reverse saturation current is nearly constant for the sufficient reverse biasing until breakdown voltage).

Current Amplification Factor (β): The ratio of change in collector current to the change in base current at constant collector emitter voltage is called the dynamic current amplification factor (β_{ac}), when the transistor is in active state. Mathematically:

$\boxed{\beta_{ac} = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}}}$ This is also known as small signal current gain and its value is fairly large (around 20 – 200).

DC Current Gain: If we simply find the ratio I_C/I_B we call the ratio as DC current gain or β_{dc}. Mathematically,

$\boxed{\beta_{dc} = I_C/I_B}$ Since the value of I_C increases almost linearly with I_B in the active state, $\frac{I_C}{I_B} \approx \frac{\Delta I_C}{\Delta I_B}$

Thus for typical applications in active state, $\boxed{\beta_{dc} = \beta_{ac} = \beta}$

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α OF A TRANSISTOR: The ratio of collector current to the emitter current is called α of a transistor. Thus mathematically,

$$\boxed{\alpha = I_C / I_E}$$

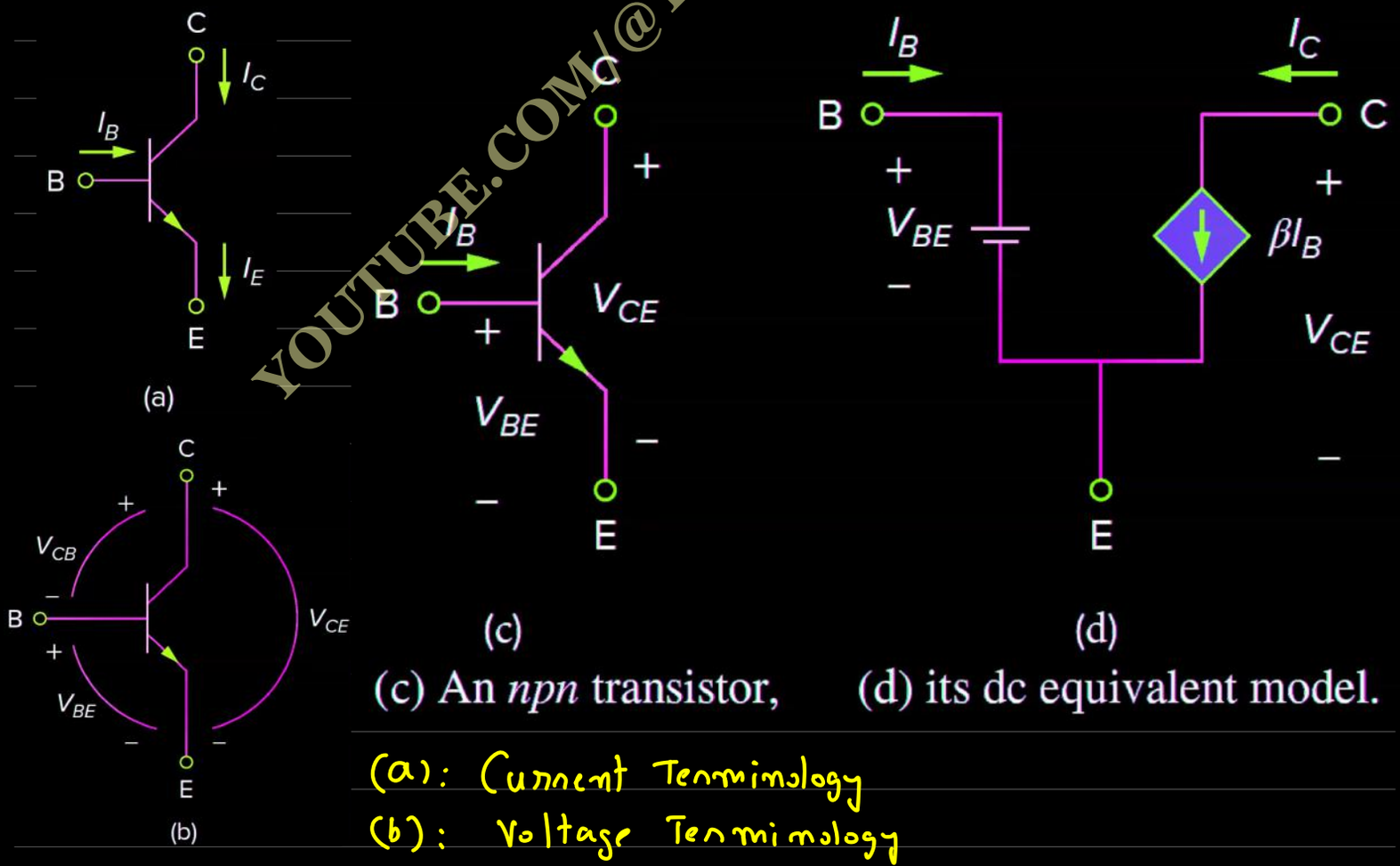
in a typical transistor application, α is very close to 1. We can easily relate α to β of a transistor. By junction law:

$$I_E = I_C + I_B \quad \Rightarrow \quad \frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta} = = I_C\left(\frac{\beta+1}{\beta}\right)$$

Thus $\boxed{\alpha = \left(\frac{\beta}{\beta+1}\right)}$ and $\boxed{\beta = \frac{\alpha}{1-\alpha}}$

TRANSFER CONDUCTANCE (TRANSCONDUCTANCE): The ratio of change in collector current to change in base emitter voltage is called transconductance. Mathematically: $\boxed{g_m = \Delta I_C / \Delta V_{BE}}$

DC MODEL OF TRANSISTORS IN ACTIVE STATE:



(c) An *npn* transistor, (d) its dc equivalent model.

(a): Current Terminology

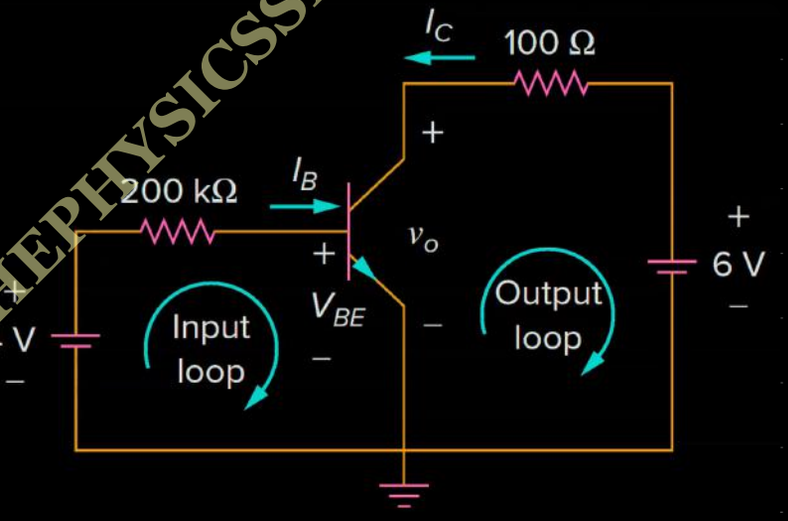
(b): Voltage Terminology

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Recall that in active state, the emitter base junction of a transistor is almost like a forward biased diode. So just like a silicon diode we can model it like a battery of 0.7 Volt against the emitter arrow (i.e current flow) in series with an ideal diode. Also, in active state, the collector current maintains a constant ratio with base current, so collector current can be simply represented by a dependent current source with current βI_B. (See figure d above). With these two simplifications, of introducing equivalent battery and a dependent current source, we can solve just about every transistor circuit.

EXAMPLE:

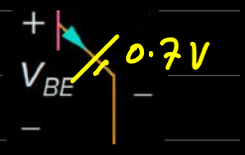
Find I_B, I_c and v_o in the shown transistor circuit. Given $\beta = 50$ and the operation is in active mode.



<u>Solution:</u> Recall that base emitter junction behaves like a 0.7 Volt battery against the emitter arrow and an ideal diode. Thus applying KVL to the input loop:

$200 \times 10^3 I_B + 0.7 - 4 = 0$ ① $\Rightarrow I_B = 16.5\,\mu A$ Ans.

(0.7 V $\uparrow V_{BE}$)



Now in the output loop, $I_c = \beta I_B = 50 \times 16.5\,\mu A = 0.825$ mA Ans.

Now KVL for output loop gives:

$$100 I_c + V_o - 6 = 0 \Rightarrow V_o = 6 - 100 I_c = 6 - 100 \times 0.825 \times 10^{-3}$$

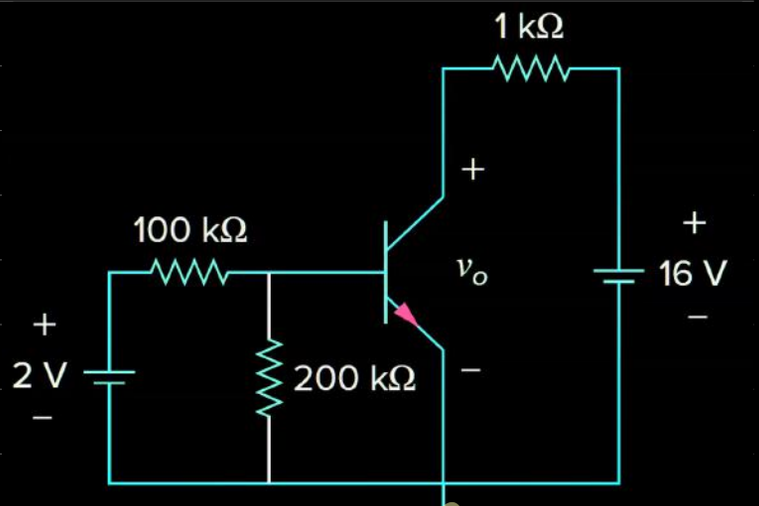
$\Rightarrow V_o = 5.9175$ Volt Ans.

(Note that here V_o is same as V_{CE}).

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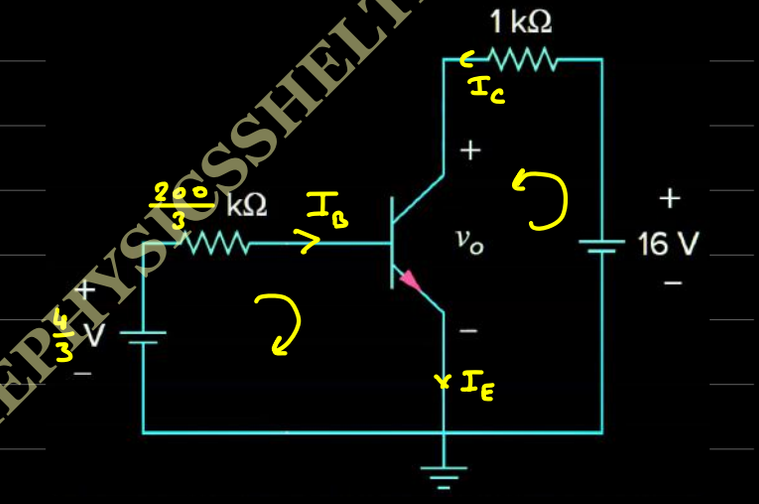
EXAMPLE:

In the shown BJT circuit, $\beta = 150$ and $V_{BE} = 0.7$ Volt. Find V_o.



Solution: Here there are three loops so either we can directly apply KVL to the three loops or alternatively we could thevenize the 2V - 100 kΩ battery branch with 200 kΩ branch and then work with two loops. Now effective millman battery: $\mathcal{E}_{eff} = \frac{2/100 + 0/200}{1/100 + 1/200}$

$\Rightarrow \mathcal{E}_{eff} = 4/3$ Volt., $R_{eff} = \frac{200}{3}$ kΩ.



Now applying KVL to the left loop:

$$\frac{2\times10^5}{3} I_B + 0.7 - \frac{4}{3} = 0$$

$\Rightarrow I_B = 9.5\,\mu A$. Now $I_C = \beta I_B = 150 \times 9.5\,\mu A = 1.425$ mA.

Now applying KVL to the right loop:

$$1.425\times10^{-3}\times1\times10^{3} + V_o - 16 = 0 \Rightarrow V_o = 16 - 1.425 = 14.575 \text{ Volt}$$

Thus $V_o = 14.575$ Volts Ans.

→ Student is advised to solve the problem without using thevenin / millman theorem also.

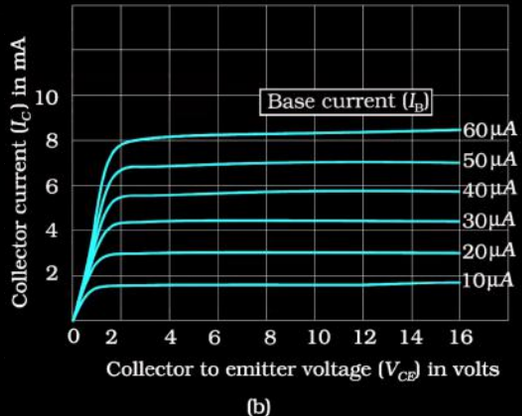
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Example 14.8 From the output characteristics shown in Fig. 14.30(b), calculate the values of β_{ac} and β_{dc} of the transistor when V_{CE} is 10 V and $I_C = 4.0$ mA.

Solution

$$\beta_{ac} = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}}, \quad \beta_{dc} = \frac{I_C}{I_B}$$

For determining β_{ac} and β_{dc} at the stated values of V_{CE} and I_C one can proceed as follows. Consider any two characteristics for two values of I_B which lie above and below the given value of I_C. Here $I_C = 4.0$ mA. (Choose characteristics for I_B= 30 and 20 μA.) At V_{CE} = 10 V we read the two values of I_C from the graph. Then



(b)

$\Delta I_B = (30 - 20)\ \mu A = 10\ \mu A$, $\Delta I_C = (4.5 - 3.0)\ mA = 1.5\ mA$

Therefore, $\beta_{ac} = 1.5\ mA / 10\ \mu A = 150$

For determining β_{dc}, either estimate the value of I_B corresponding to $I_C = 4.0\ mA$ at $V_{CE} = 10\ V$ or calculate the two values of β_{dc} for the two characteristics chosen and find their mean.

Therefore, for $I_C = 4.5\ mA$ and $I_B = 30\ \mu A$,

$\beta_{dc} = 4.5\ mA / 30\ \mu A = 150$

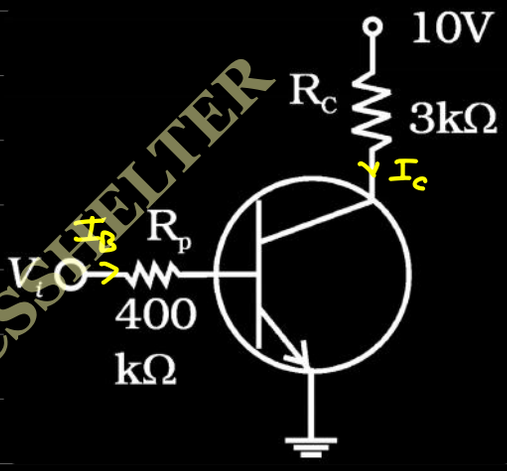
and for $I_C = 3.0\ mA$ and $I_B = 20\ \mu A$

$\beta_{dc} = 3.0\ mA / 20\ \mu A = 150$

Hence, $\beta_{dc} = (150 + 150)/2 = 150$

EXAMPLE: NCERT EXEMPLAR 14.32:

In the shown circuit when the input voltage of the base resistance is 10V, $V_{BE} = 0$ and V_{CE} is also zero. Find the value of I_B, I_C and β.



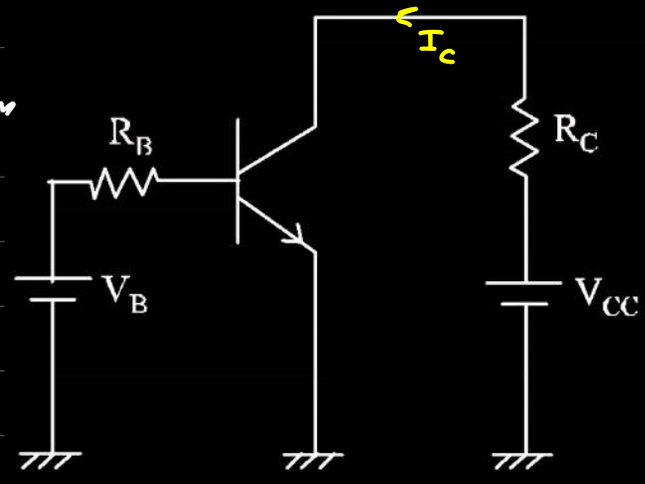
Solution: Imagine a battery of V_i between earth and V_i, then in the input loop,

$I_B \times 400K - V_i = 0 \Rightarrow I_B = V_i/400K = \frac{10}{400K} = 25\ \mu A$. Now imagine a battery of 10 Volts between earth and collector terminal. KVL for output loop gives us: $3k \times I_C + V_{CE} - 10 = 0 \Rightarrow I_C = 10/3k = 3.33\ mA$.

Now $\beta = I_C/I_B = 3.33\ mA / 25\ \mu A \approx 133$ Ans.

EXAMPLE: JEE MAIN 2019:

In the shown circuit $\beta = 250$, $R_C = 1k\Omega$ and $V_{CC} = 10$ Volts. What is the minimum base current for V_{CE} to reach saturation.



Solution: $V_{CE} = V_{CC} - I_C R_C$. For saturation, $V_{CE} = 0$ also $I_C = \beta I_B$, thus,

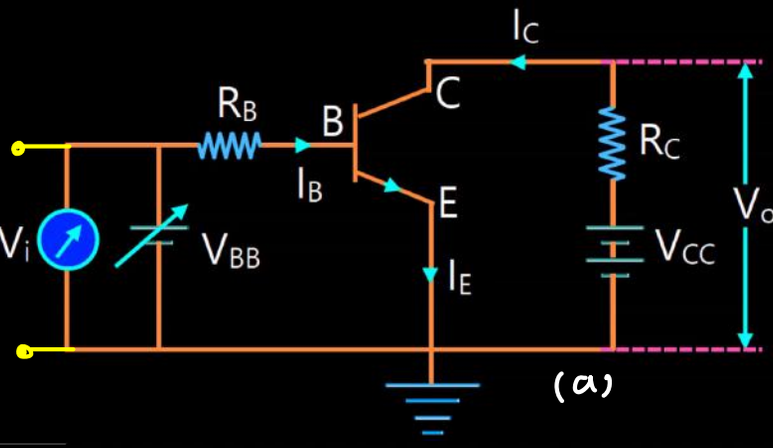
$V_{CC} - \beta I_B R_C = 0 \Rightarrow I_B = V_{CC}/\beta R_C$

$\Rightarrow I_B = 10/(250 \times 1000) = 40\ \mu A$ Ans.

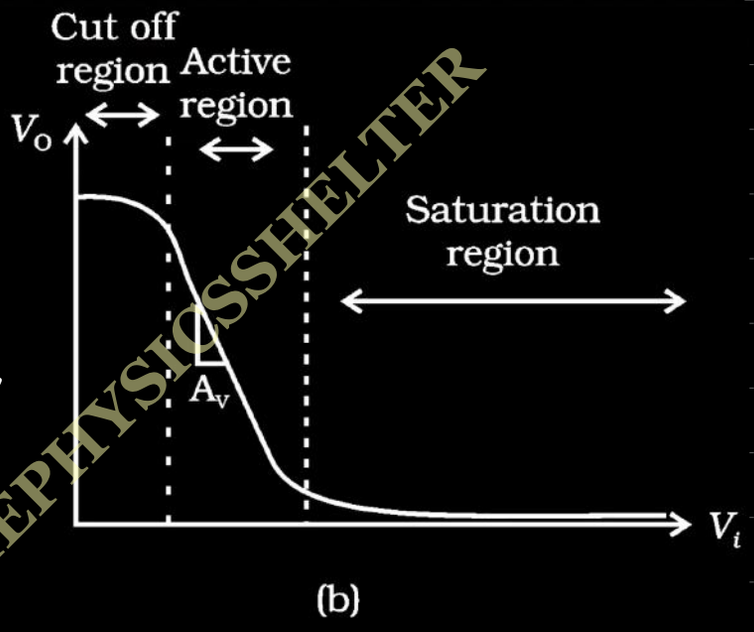
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TRANSISTOR AS SWITCH :

A switch is a device which generates a high output voltage for a low input voltage and vice versa. This can be easily achieved using a transistor circuit as shown in the figure. The output voltage is given as $V_o = V_{cc} - I_c R_c$, now when input voltage is very high, base emitter voltage is sufficient to produce a large base current and hence a large collector current accordingly transistor goes in saturation and the output voltage becomes low. (See the adjacent curve from NCERT). Also when input voltage is very low, the emitter is not properly biased thus leading to a negligible collector current. Accordingly $V_{cc} - I_c R_c \approx V_{cc}$. Thus corresponding to a low biasing voltage on the input side, output voltage is high because the transistor is operating in cut-off region. Thus this operation of transistor involving use of cut off and saturation regions is called the use of transistor as a switch (or a NOT gate).



(a)

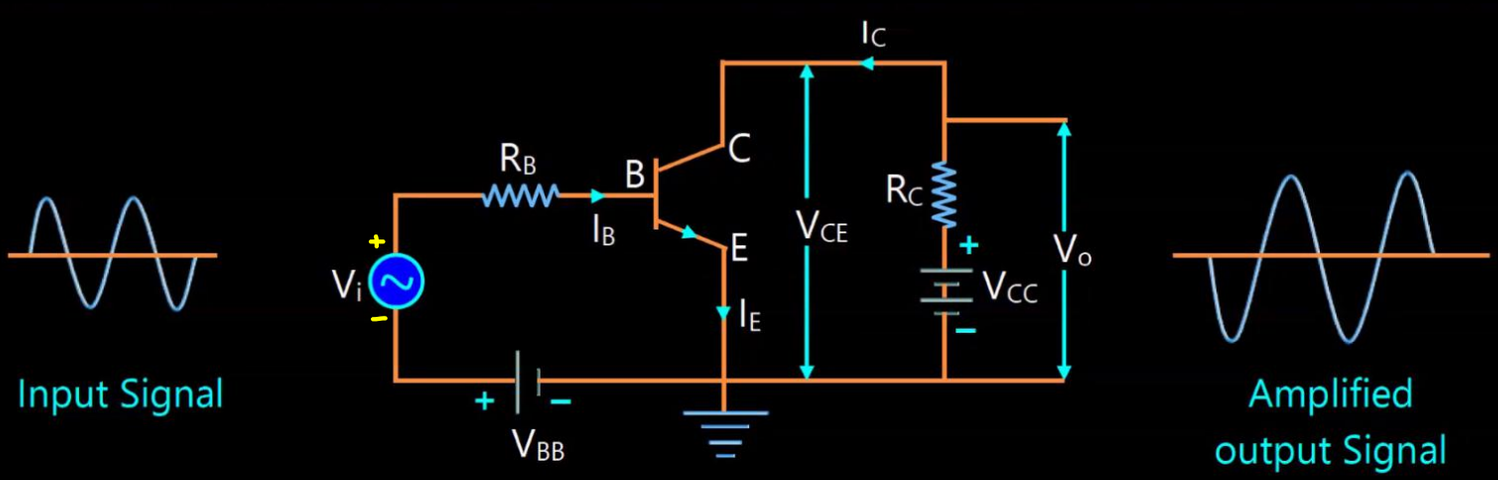


(b)

→ Also see the output vs input graph in active region. For sufficiently large value of R_c, this curve will have a very large slope. Thus a small change in input voltage will bring about a large change in output. This forms the basis of using transistor as an amplifier about which we will study in more detail in the next section.

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Transistor As Common Emitter Amplifier:



Common Emitter Amplifier NPN Transistor

Suppose we wish to amplify an audio signal coming from a microphone (i.e increase a low voltage fluctuation to a high voltage fluctuation which drives a speaker). In the circuit shown above, consider V_i as a voltage fluctuation generated by a microphone and V_o as the voltage used to drive a speaker. We will show how V_o will be an amplified version of V_i (with a π shift) with the given circuit working in active mode. Now on the input side we can write:

$$-V_i + I_B R_B + V_{BE} - V_{BB} = 0 \Rightarrow \Delta V_i = R_B \Delta I_B \quad (1)$$

($\because$ $V_{BE} = 0.7$ Volts $\approx$ constant and V_{BB} is also constant).

Now for the output loop: $V_o = V_{cc} - I_c R_C \Rightarrow \Delta V_o = -R_c \Delta I_c$ (2)

Now Voltage gain of an amplifier is defined as $A_v = \frac{\Delta V_o}{\Delta V_i}$, thus

using equations (1) & (2), $A_v = -\frac{R_C}{R_B} \cdot \frac{\Delta I_C}{\Delta I_B}$, but $\frac{\Delta I_C}{\Delta I_B} = \beta_{ac} \approx \beta_{dc} = \beta$

thus $\boxed{A_v = -\beta \frac{R_C}{R_B}}$ ← Voltage gain of Common emitter Amplifier.

→ Now using suitable value of R_c and R_B we can achieve the desired voltage gain.

→ Notice the negative sign. This means output voltage is 180° out of phase with input voltage (also see the figure at the top of the section).

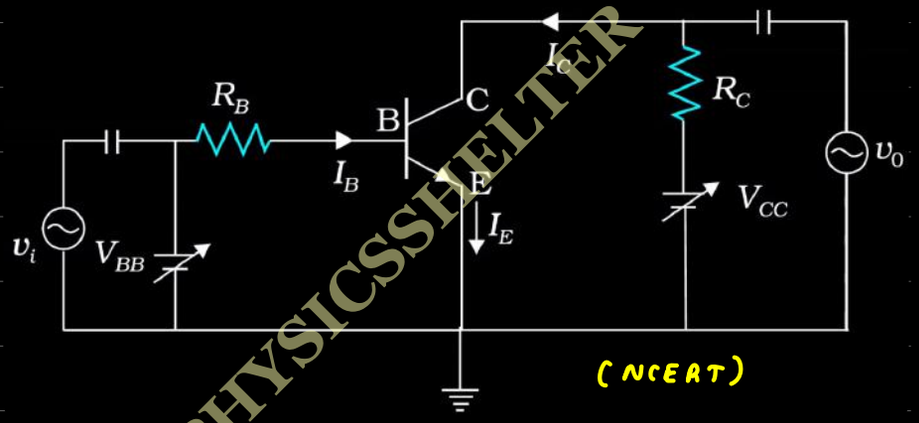
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POWER GAIN: The product of voltage gain and current gain is called power gain. Thus, $A_p = A_v \times \beta$ thus: $A_p = -\beta \frac{R_C}{R_B} \times \beta$ or:

$$\boxed{A_p = -\beta^2 R_C/R_B}$$ → The transistor does not generate power the power gain happens at the expense of work done by the batteries.

AMPLIFIER WITH CAPACITOR ISOLATORS:

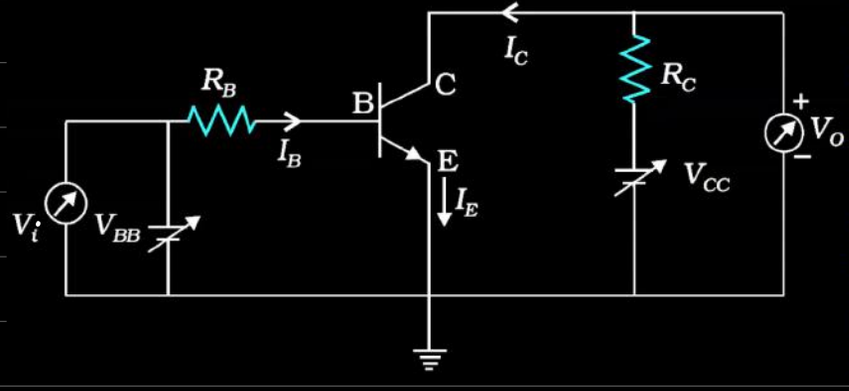
For easy understanding of Amplifier circuits we used a simplified version of the amplifier circuit. In practice we are interested only in fluctuating (A.C.) components of the input and output signals. A capacitor can block out dc and allow only a.c. to pass accordingly amplifier circuits use capacitors for filtering out dc. One such circuit is shown above. Its working is not dealt at JEE level (you will learn it in electronics engineering courses). For our purposes we will just replace the circuit by our simplified version.



(NCERT)

EXAMPLE: NCERT

Example 14.9 In Fig. 14.31(a), the V_{BB} supply can be varied from 0V to 5.0 V. The Si transistor has β_{dc} = 250 and R_B = 100 kΩ, R_C = 1 KΩ, V_{CC} = 5.0V. Assume that when the transistor is saturated, V_{CE} = 0V and V_{BE} = 0.8V. Calculate (a) the minimum base current, for which the transistor will reach saturation. Hence, (b) determine V_i when the transistor is 'switched on'. (c) find the ranges of V_i for which the transistor is 'switched off' and 'switched on'.



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Solution: (a) For saturation, $V_{CE} = 0 \Rightarrow V_{CC} - I_C R_C = 0 \Rightarrow V_{CC} - \beta I_B R_C = 0$

$\Rightarrow I_B = V_{CC} / \beta R_C = 5/(250 \times 10^3) = 20\,\mu A$ Ans.

(b) Now $V_i = I_B R_B + V_{BE} = 20\,\mu A \times 100\,k\Omega + 0.8 = 2.8$ Volt Ans.

→ Switched "ON" means saturation. So transistor remains switched on from 2.8 Volts to 5.0 Volts.

(c) For transistor to be cut off, the base emitter voltage should be insufficient to create proper forward bias. Thus $V_i < 0.6$ Volts.

Thus for switched off state, $0 < V_i < 0.6$ Volt and for switched on state, $2.8 < V_i < 5.0$ Volt.

→ Note that transistor in active state for $0 < I_B < 20\,\mu A$ this corresponds to $0.6 < V_i < 2.8$ Volts.

→ $I_C = \beta I_B$ holds only in active state (or just till verge of achieving saturation) after that $I_C < \beta I_B$

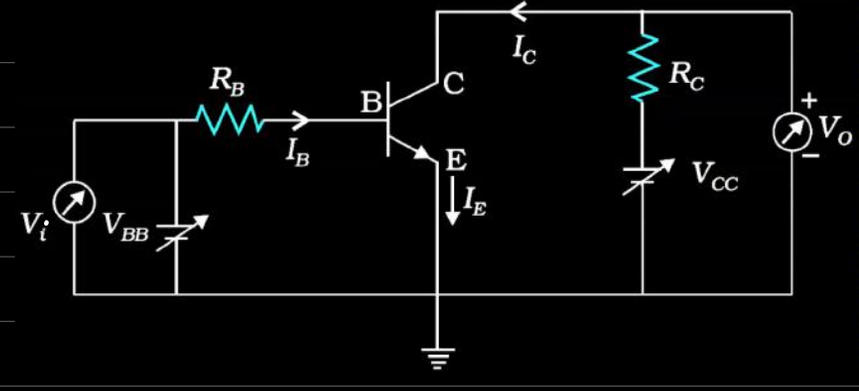
Example 14.10 For a CE transistor amplifier, the audio signal voltage across the collector resistance of 2.0 kΩ is 2.0 V. Suppose the current amplification factor of the transistor is 100, What should be the value of R_B in series with V_{BB} supply of 2.0 V if the dc base current has to be 10 times the signal current. Also calculate the dc drop across the collector resistance. (Refer to Fig. from previous example).

Solution The output ac voltage is 2.0 V. So, the ac collector current i_C = 2.0/2000 = 1.0 mA. The signal current through the base is, therefore given by $i_B = i_C / \beta$ = 1.0 mA/100 = 0.010 mA. The dc base current has to be 10× 0.010 = 0.10 mA.

From input loop , $R_B = (V_{BB} - V_{BE}) / I_B$. Assuming V_{BE} = 0.6 V,

R_B = (2.0 – 0.6)/0.10 = 14 kΩ.

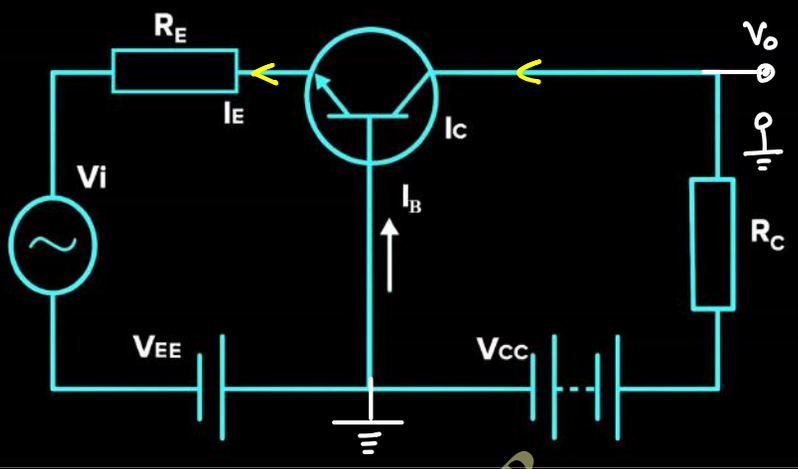
The dc collector current I_C = 100×0.10 = 10 mA.



dc drop across R_C = 10mA × 2kΩ = 20 V

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Transistor As Common Base Amplifier:



The adjacent figure shows circuit for common base amplifier.

Here $V_i = -I_E R_E - V_{BE} + V_{EE}$

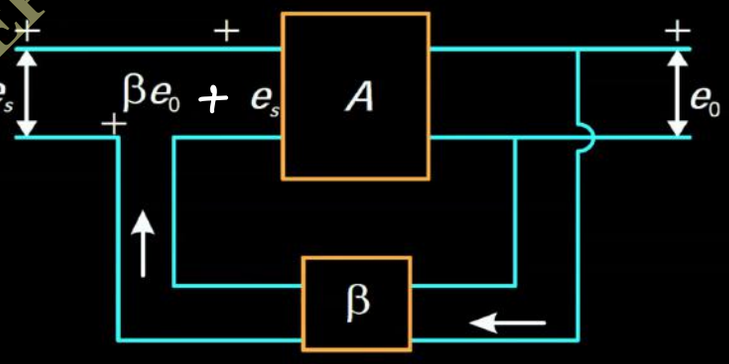
$\Rightarrow \Delta V_i = -R_E \Delta I_E$ (1) now on output side, $V_o = V_{cc} - I_c R_C$

$\Rightarrow \Delta V_o = -R_C \Delta I_c$ thus voltage gain:

$A_V = \Delta V_o / \Delta V_i = \frac{\Delta I_E}{\Delta I_c} \cdot \left(\frac{R_E}{R_C}\right)$. Here we see that there is no negative sign.

→ In a common base amplifier, input and output are in same phase.

Concept Of Feedback In Amplifiers:



When a part of the output voltage (or current) is superposed on injected back in to the input circuit after being multiplied by some factor (β) we say that it is a circuit with feedback. If the feedback voltage is in phase with input we call it positive feedback or regenerative feedback. However if feedback is in opposite phase to the input we call it negative feedback, or degenerative. To understand the mathematics of feedback circuits, consider the schematic circuit in the figure. Here, e_s: source voltage, e_o: output voltage, e_f: feedback voltage (i.e. βe_o) e_i: net input voltage (i.e. $e_s + \beta e_o$) A: Amplification factor of the amplifier. Now:

Net input voltage: $e_i = e_s + e_f = e_s + \beta e_o$, thus output voltage, $e_o = A(e_s + \beta e_o)$, now rearranging this equation we get:

$$e_o = \frac{e_s A}{1 - \beta A} \Rightarrow \text{Effective amplification, } A_f = \frac{e_o}{e_s} = \frac{A}{1 - \beta A}$$

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thus $\boxed{A_f = \frac{e_o}{e_s} = \frac{A}{1-\beta A}}$ Normally in a transistor amplifier, for output, we need some input signal however from the effective amplification equation we can see that if $\beta = 1/A$, the gain of amplifier becomes infinite and an output can be obtained without any input signal. This forms the basis of feedback oscillators which continue to generate alternating voltage signals without any input signal. ($\beta A = 1$ or $|\beta A| = 1$ is called Barkhausen criterion for sustained oscillations). (With negative feedback module writes $A_f = A/(1+\beta A)$, so keeps β unsigned).

TRANSISTOR AS OSCILLATOR:

An oscillator generates an alternating voltage from d.c. batteries. One of the ways of generating A.C. voltages from dc could be using the LC circuits, however invariably every circuit has some resistance and therefore the LC circuit would undergo damped oscillations and the oscillations will die out. This is taken care of by the feedback oscillators. One such circuit is shown in adjacent figure. Here all the three inductor coils (T_1, T_2 and T_2') are wound on same iron core and are magnetically coupled. When switch S_1 is closed, current through the inductor T_2 rises thus increasing current across it.

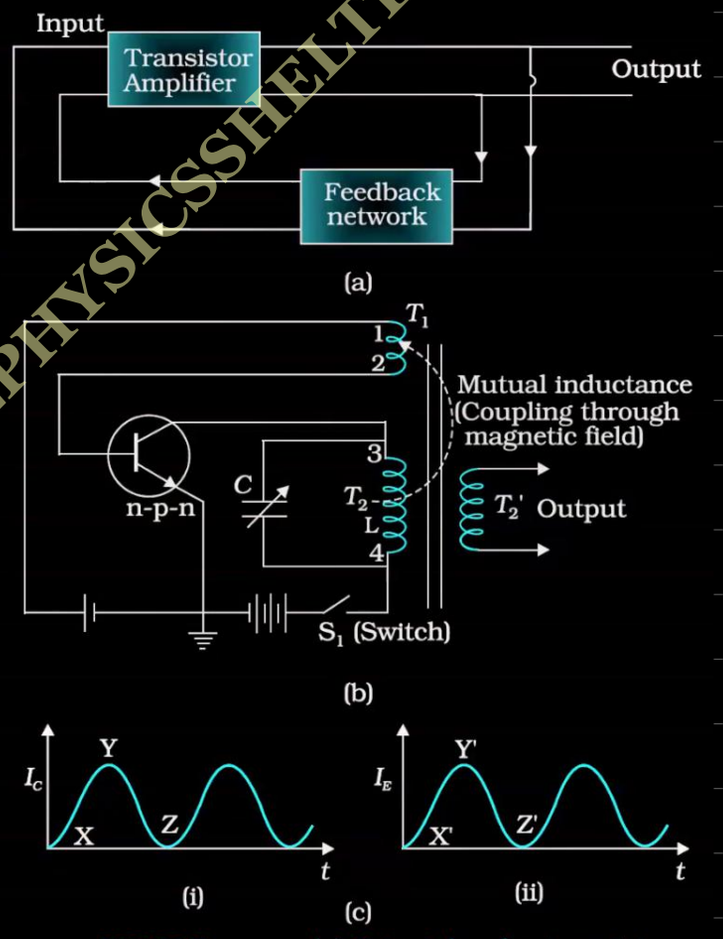


FIGURE (NCERT) (a) Principle of a transistor amplifier with positive feedback working as an oscillator and (b) Tuned collector oscillator, (c) Rise and fall (or built up) of current I_c and I_e due to the inductive coupling.

Now since T_2 is coupled with T_1 it induces a voltage in it (positive feedback), thus initially increasing the effective base emitter bias. Note that this feedback voltage is governed by the slope of the current and not the current value itself ($\because \mathcal{E} \propto \frac{dI}{dt}$). Due to positive feedback collector current increases to saturation (maximum current) and the feedback voltage becomes zero. After this the

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collector current begins to decrease causing a reduction of voltage on the base emitter junction which causes emitter to emit lesser and lesser majority carriers thus collector current continues to reduce until the transistor reaches cut-off state due to continuously reducing biasing voltage on emitter side. Now one cycle is complete and we are back to the initial configuration.
In this manner the circuit alternates between saturation and cut-off states thus producing alternating voltages.

→ The frequency of oscillation is governed by the tuned circuit on the tank circuit (i.e L and C in the given circuit). The resonance frequency of the tank circuit governs the oscillation frequency, therefore such a circuit is called tuned oscillator. The frequency of oscillation is given as $\boxed{\nu = \frac{1}{2\pi\sqrt{LC}}}$

→ Here our tuned circuit was on the collector side that's why it is called tuned collector oscillator if it were on the base side, we would call it tuned base oscillator.

→ There can be alternative types of tank circuits (say RC circuit), depending on the design of tank circuit, the oscillator may be named differently (for example Colpitt oscillator, Hartley oscillator and RC oscillator etc).

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EXAMPLE: In a negative feedback amplifier, the gain without feedback is $A = 100$. The feedback ratio $(\beta) = 1/25$ and the input voltage is 50 mV. Calculate (a) Gain with feedback (b) feedback factor (c) Output voltage (d) Feedback voltage (e) New input voltage so that so that output voltage with feedback becomes equal to output voltage without feedback. (Feedback is negative)

Solution: $A_f = \frac{A}{1-\beta A}$, $\beta = -1/25$, $A = 100 \Rightarrow A_f = \frac{100}{1-(-1/25 \times 100)}$

$\Rightarrow A_f = 20$. (b) $|\beta| = 1/25$, $\Rightarrow \beta = -1/25$ (module considers $\beta = 1/25$ and modifies the formula for negative feedback as $A_f = A/(1+\beta A)$. This is just a matter of convention). (c) $V_0 = A_f V_i = 50\,mV \times 20$
$\Rightarrow V_0 = 1$ Volt Ans. (d) $V_f = \beta V_0 = \frac{-1}{25} \times 1 = -1/25$ Volt Ans.

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(Module gives +1/25 Volt again a convention matter). (e) Let V_n be new required input voltage to get same output as one without feedback then: $V_n \times A_f = V_i \times A \Rightarrow V_n = V_i \times \frac{A}{A_f} = 5V_i$.

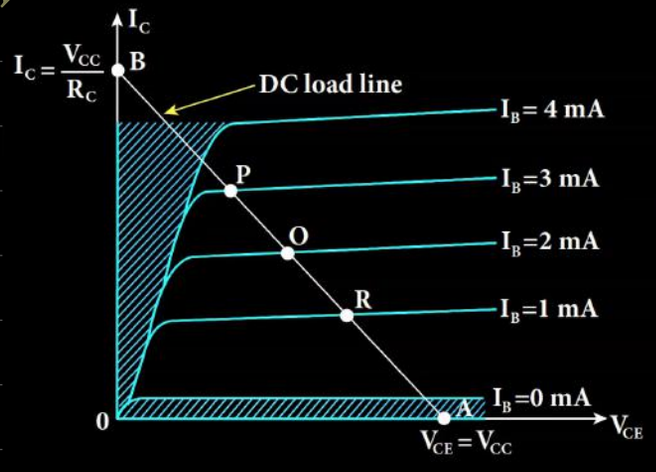
Thus $V_n = 50mV \times 5 = 250\ mV$ Ans

EXAMPLE: Calculate the range of variable capacitor to be used in a tuned oscillator with fixed inductance $150\mu H$. The frequency band is from 500 kHz to 1500 kHz.

Solution: $\nu = \frac{1}{2\pi\sqrt{LC}} \Rightarrow C = \frac{1}{4\pi^2\nu^2 L}$, now we just need to substitute the bounds of required frequency to get the range of variable capacitor. Substituting the values we get C lying between 75pF and 676 pF Ans.

DC LOAD LINE AND OPERATING POINT:

The operating point is a point where the transistor can be operated efficiently. A line that is drawn with the values V_{CC} (with $I_C = 0$) and I_C (when $V_{CE} = 0$) is called the dc load line. The dc load line is superposed on transistor output characteristics and the intersections obtained are called quiescent points (P, Q, R are Quiscent points (or Q points) in the figure. For operation as an amplifier, generally Q point is chosen at the middle of the load line and for using as switch, operating points may be chosen in cut off or saturation region on the load line.



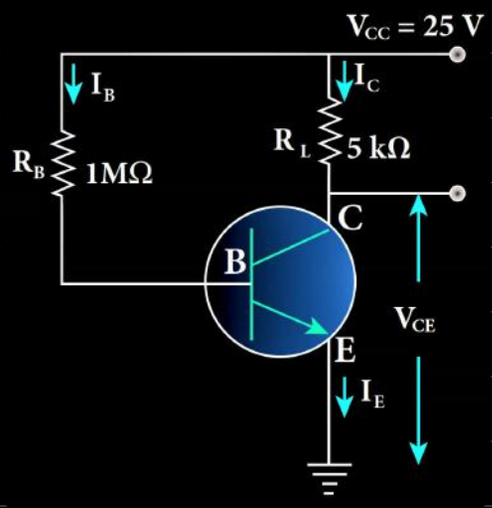
Output characteristics of a transistor in common emitter mode with the dc load line

→ Choosing Q at the middle is just a design thumbrule and not a hard and fast recommendations. In actual problems, Q point could be away from mid-point also.

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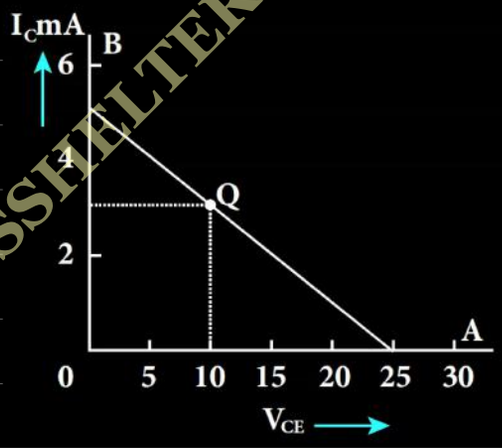
EXAMPLE:

The current gain of a common emitter transistor circuit shown in the figure is $\beta = 120$. Draw the dc load line and mark the Q point corresponding to the given circuit parameters. Ignore V_{BE}. (i.e take $V_{BE} = 0$).



Solution:

The equation of the load line is given as $V_{CE} = V_{CC} - I_C R_C$, for our case $V_{CC} = 25$ Volt, So the load line is as shown in the figure. Now we need to locate the Q point on the load line.



Now using node voltage, $I_B = (25-0)/1\times10^6$ A $\Rightarrow I_B = 25\,\mu A$, Now $I_C = \beta I_B = 120 \times 25\,\mu A = 3$ mA. The corresponding value of V_{CE}: $V_{CC} - I_C R_C = 25 - 3\text{mA} \times 5\text{k}$ Volt $= 10$ Volts. Thus our required Q point is located at (10 Volt, 3mA) Ans.

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DIGITAL ELECTRONICS:

Digital electronics is a sub-branch of electronics which deals with digital (or binary) signals. A binary signal has only two possible states: high and low (or 0 and 1 or True and false). It is possible to represent all kinds of data by encoding it in binary signals. All computers store the data using this technique of encoding all the data in to strings of zeroes and ones.

BINARY COUNTING:

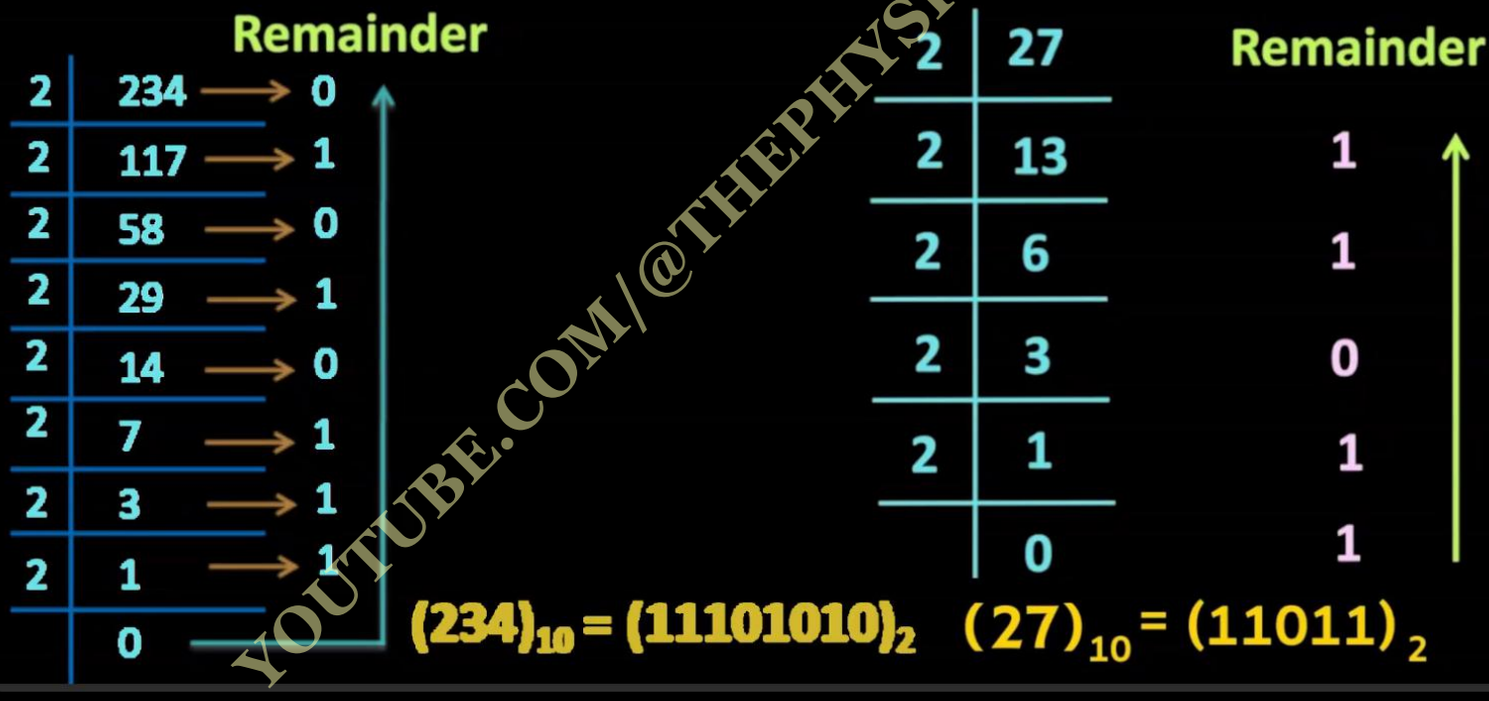
Just as usual counting system relies on 10 symbols (0 → 9) binary counting system relies on just two symbols (0 and 1). Thus, Some numbers and their binary equivalents can be listed as shown in the table on the following page.

Notice that here 1 behaves somewhat like the behavior of 9 in the usual decimal system for instance after 11 we get 100 in binary just as we get 100 after 99 in decimal system. In decimal system place value of any number is obtained by multiplying the face values by appropriate powers of 10, similarly here the place values are obtained by multiplying by powers of 2.

Decimal Number	Binary Number
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010

For instance $10110)_2$ is $1\times2^5 + 0\times2^4 + 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 = 45$.

Decimal To Binary Conversion:



To convert any number to binary, it is repeatedly divided by two and the remainder is noted down at every step (see figures above) Now the binary equivalent is given by reverse sequence of remainders as illustrated above.

→ The same method can be used to convert the number to any base (Most popular bases are binary, Oct (base 8) and Hex (Base 16).

Base 16 uses 6 additional symbols after 9 i.e A, B, C, D, E F, for instance a remainder of 12 on division by 16 will be written as C).

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EXAMPLE: Convert 23 to binary number.

Solution: We apply the algorithm of repeated division and take the reverse sequence of remainders. This is illustrated in adjacent figure. Thus $23_{10} = 10111_2$ Ans.

2	23
2	11 – 1
2	5 – 1
2	2 – 1
2	1 – 0
2	0 – 1

EXAMPLE:

Convert 1101011_2 to decimal number.

Solution: $1\times2^0 + 1\times2^1 + 0\times2^2 + 1\times2^3 + 0\times2^4 + 1\times2^5 + 1\times2^6 = 107$ Ans.

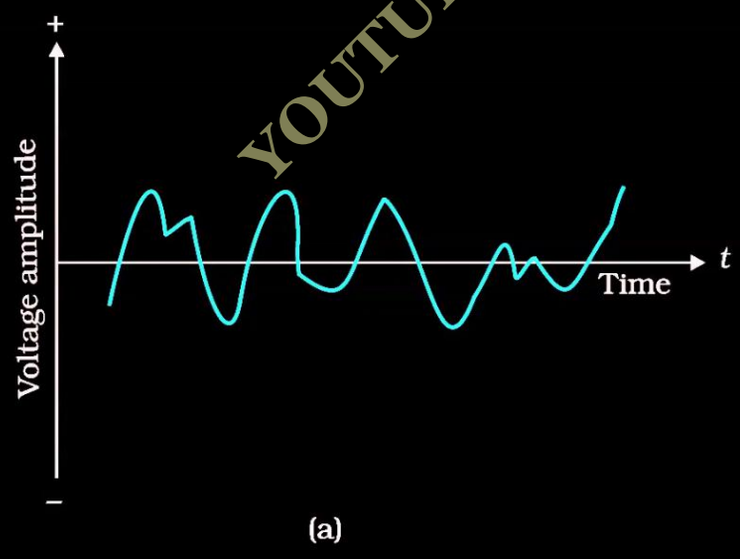
EXAMPLE: Convert 107_{10} to Hex.

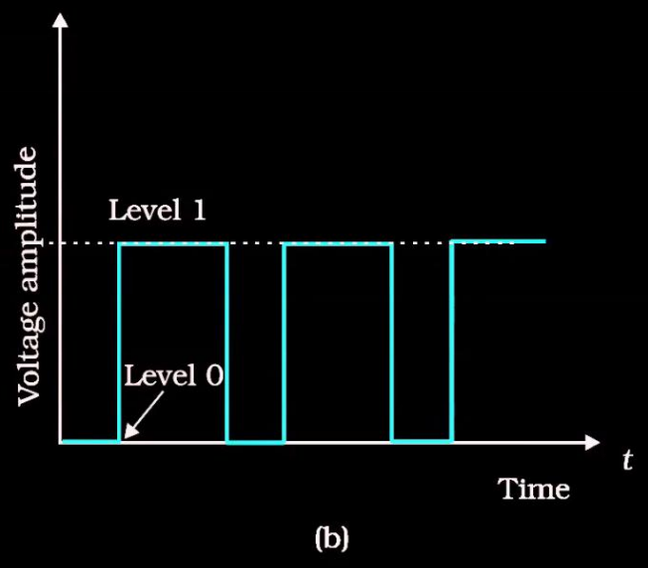
Solution: The process is shown in adjacent column. Using algorithm of repeated division and taking remainders from 16, $107_{10} \equiv 6b_{16}$ Ans.

→ Similarly we can show that $107_{10} \equiv 153_8$.

16	107
16	$6 - 11 \equiv b$
	$0 - 6 \equiv 6$

ANALOG AND DIGITAL SIGNALS:





(a) Analogue signal, (b) Digital signal.

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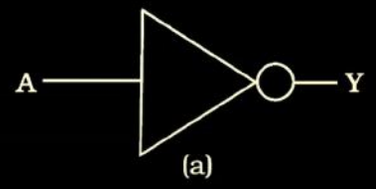
A signal represents coded information for example when we speak in to a microphone, it converts our voice in to voltage or current signals which can be processed to generate audio output. When signals can take continuous set of values (such as microphone voltage output), it is called an analog signal. Digital signals are the signals which take only binary values (i.e 0 or 1). The information in the digital signals is codes as strings of zeroes and ones which can be appropriately processed to generate the required output in a usable form. (For instance a video file on a computer is encoded as strings of zeroes and ones and they can be processed to generate colors and audio outputs in form of a movie playing on the screen). For illustration, an example of digital and analog signals is presented at the beginning of this section.

LOGIC GATES:

A digital circuit having some relationship between input and output voltages in terms of their digital values (High / low) is called a logic gate. Various logic gates that are used in digital electronics are described in the following sections.

NOT GATE: (INVERTER): ($\bar{}$ OPERATOR)

This is a logic gate whose output is just the inverse of the input i.e a high input generates a low output and a low input generates a high output. The symbol and truth table for NOT GATE is shown in adjacent figures. The symbol for boolean 'NOT' operation is $\bar{\square}$, for instance here we can write the boolean equation in terms of the NOT operator as $Y = \bar{A}$.
(A truth table is an exhaustive table of all possible inputs and corresponding outputs in a logic circuit).



(a)

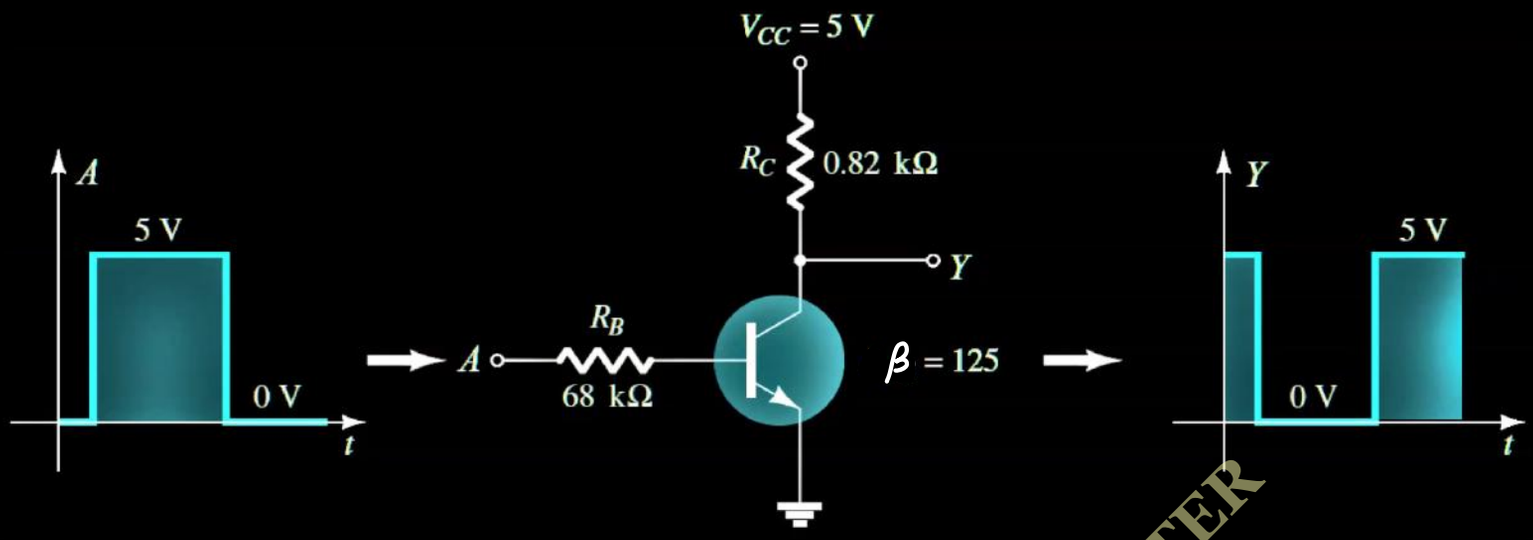
Input	Output
A	Y
0	1
1	0

(b)

(a) Logic symbol,
(b) Truth table of NOT gate.

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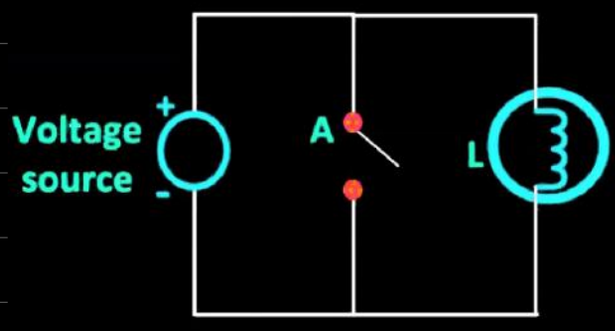
Realization Of A Not Gate Using Transistor:



The above figure illustrates how a transistor can be used to realize a NOT gate. Here A is the input (Base-Emitter biasing voltage) and Y is the output. When A is high, the transistor will go in to saturation and accordingly Y (which is V_{CE}) will become zero (recall that, $V_{CE} = V_{CC} - I_C R_C$ which goes to zero if current is large enough). Now when input is low, emitter does not emit properly and transistor is cut-off. Accordingly $I_C = 0$ and $Y = V_{CC} - I_C R_C = V_{CC} = 5$ Volts as high. Thus our transistor circuit does the job of a NOT Gate.

Realizing A Not Gate Using Switching Circuit:

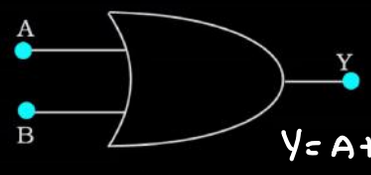
If a boolean value of 1 is assigned to the switch being closed (considered as input) and a boolean value of 0 is assigned to open switch then if glow of bulb is considered as 1 (output) one can readily see that adjacent circuit acts like a not gate according to the truth table.



When switch is closed, the bulb does not glow due to being short circuited and when the switch is open, the bulb glows due to power supply from voltage source.

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OR GATE: (+ OPERATOR)

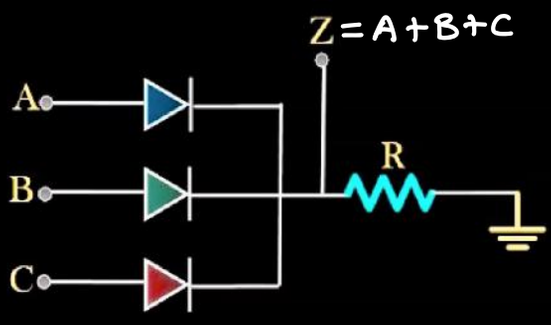


(a)

Input		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

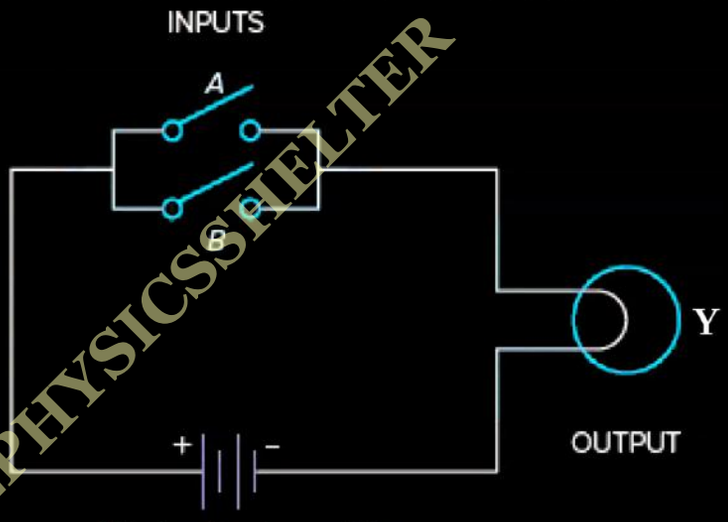
(b)

(a) Logic symbol (b) Truth table of OR gate.



Circuit diagram of OR gate using Diodes

An OR gates has two or more inputs and one output. Its symbol and truth table is shown in the figure above. An OR gates outputs a 1 if any of its inputs becomes 1. The output of zero is achieved only when all inputs are simultaneously zero.



OR circuit using switches.

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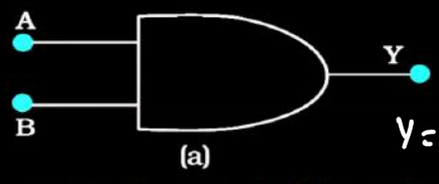
REALIZING OR GATE USING DIODES:

Schematic circuit for realizing OR Gate using diodes is shown above. It can be readily seen that if any of the three inputs A, B or C are made high then the diode will behave like short circuit due to being forward biased and Z will become high. When all the three inputs are low, there is no current through the resistor and accordingly Z gets same potential as earth (i.e. low output).

REALIZING OR GATE USING SWITCHING CIRCUIT:

Schematic circuit for realizing OR gate using switching circuits is presented in one of the figures above. It can be clearly seen that if any one of the two switch A or B are closed (Boolean input 1), the bulb glows (boolean output 1).

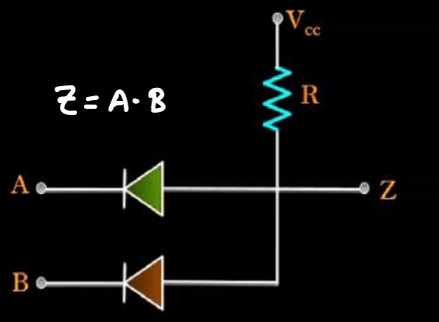
AND GATE: (· OPERATOR):



(a) Logic symbol, (b) Truth table of AND gate.

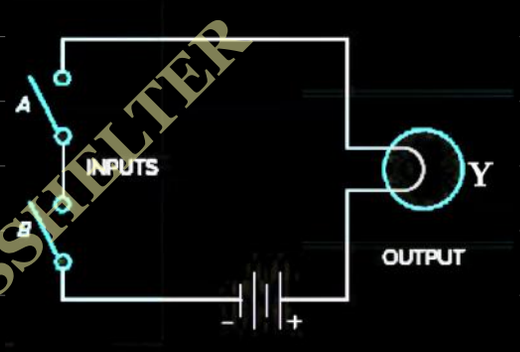
Input		Output
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(b)



Circuit diagram of AND gate using Diode

An AND gate has two or more inputs and one output. The output Y of the AND gate is 1 only when input A and input B are both 1. The symbol and truth table of AND gate is shown in the figure above.



AND circuit using switches.

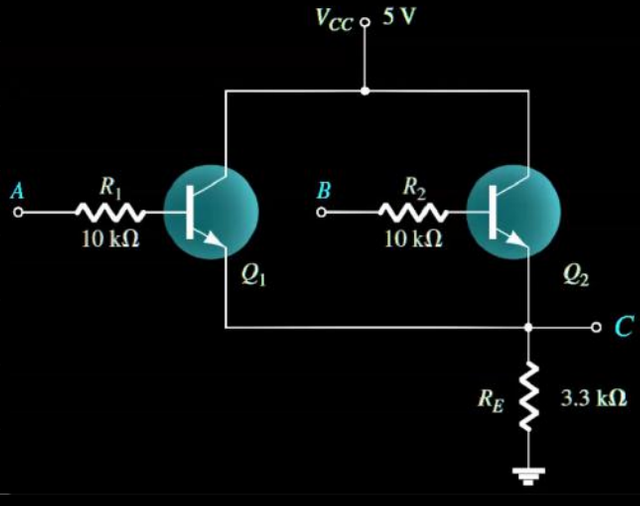
REALIZATION OF AND GATE USING DIODES:

The schematic circuit for realizing an AND gate using diodes is shown in one of the figures above. Here if any of the inputs A or B are low, that diode becomes short circuited and accordingly, Z gets shorted with low potential (Thus low output). When both inputs are high, both diodes behave like an open circuit and thus no current flows through the resistor. Since there is no drop across the resistor, the potential of Z becomes same as that of V_{cc}.

AND GATE USING SWITCHING CIRCUIT: See the switching circuit figure. When both A and B are closed, bulb gets lit (1), it remains off otherwise (0).

EXAMPLE:

Which Gate is represented by the circuit shown in adjacent figure. Here A and B are inputs and C is the output.

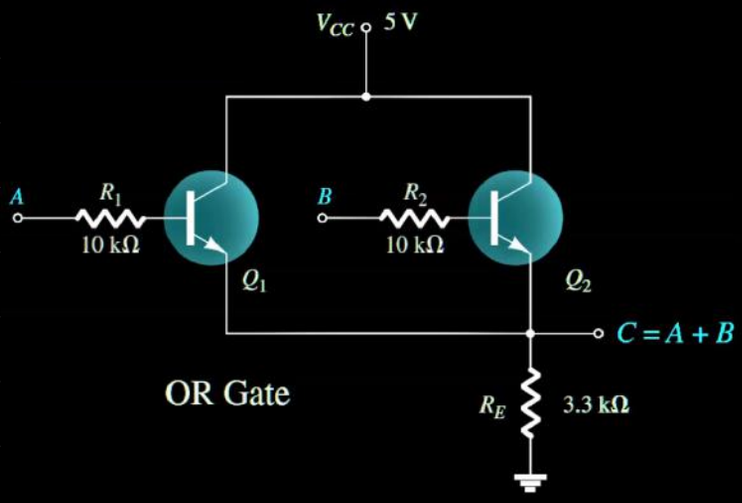


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Solution:

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

1 = high
0 = low

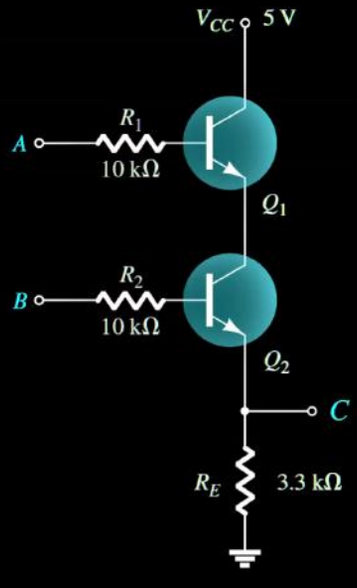


OR Gate

Here if any of the inputs A or B become high, the corresponding transistor will go in to saturation thus causing V_{CE} becoming zero across it, thus shorting V_{cc} with point C and hence causing C to be at high potential. If both inputs are low, then both transistors are cut off and no current can flow through their emitters. Thus current through R_E will become zero and hence C will be at low potential. Looking at these facts we prepare a truth table and the circuit works like an OR gate

EXAMPLE:

Identify the gate represented by the transistor circuit shown in adjacent figure. Here A and B are inputs and C is the output.

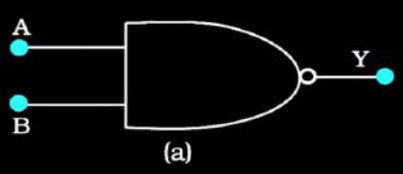


Solution: Here for C to be shorted with V_{cc}, V_{CE} across both transistors must be zero i.e both transistors must be in saturation mode. Thus to get high output at C, both inputs should be high. This is characteristic of AND gate. Thus this circuit represents AND gate.

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NAND GATE :

$Y = \overline{A \cdot B}$

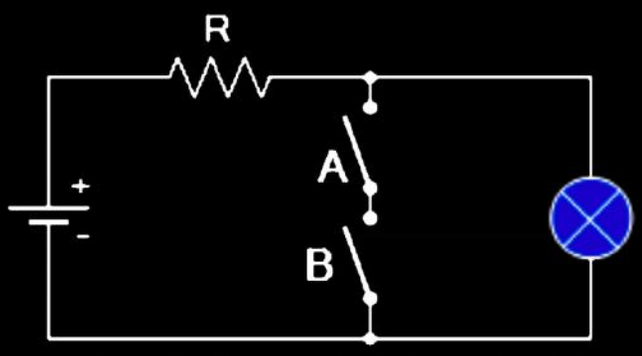


(a)

Input		Output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

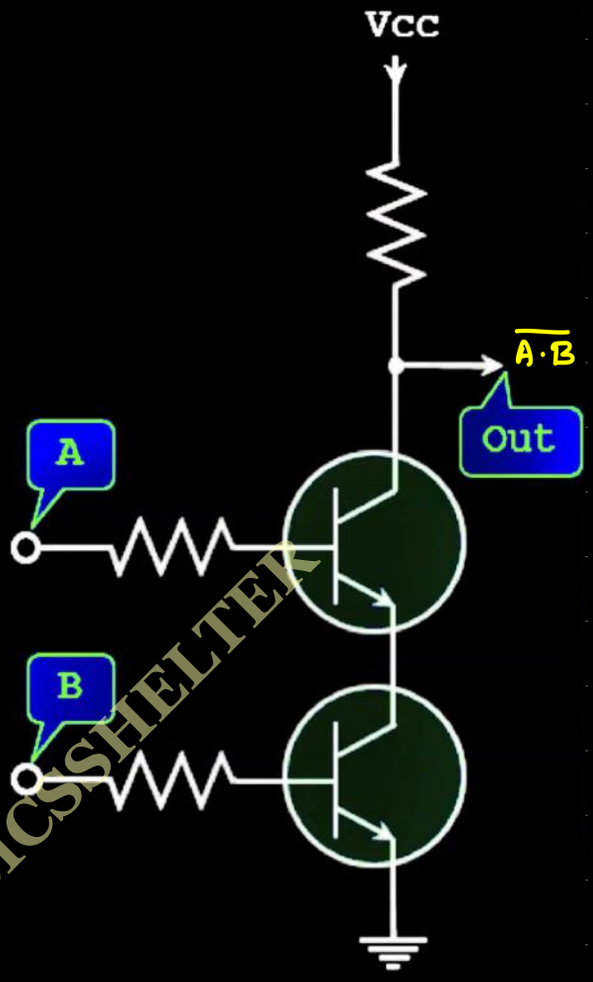
(b)

(a) Logic symbol, (b) Truth table of NAND gate.



Lamp - ON = "1"
Lamp - OFF = "0"

Switch A - Open = "0", Closed = "1"
Switch B - Open = "0", Closed = "1"



A NAND gate is an AND gate followed by a not gate. Its symbol and the truth table are shown in the figure above. Its output is exactly the binary inverse of that of AND gate, i.e it gives a zero only when both inputs are 1, in every other case its output is zero.

REALIZATION OF NAND GATE USING TRANSISTORS :

Consider the transistor circuit shown in the figure above. Here when both inputs are high, both transistors go in to saturation, and V_{CE} becomes zero. Thus output becomes effectively shorted with ground and thus becomes zero. If any of the inputs A and B is low, current from V_{cc} can't flow to the ground and thus output voltage is same as V_{cc}. Thus using this circuit we get the desired truth table for NAND gate.

REALIZATION OF NAND GATE USING SWITCHING CIRCUITS :

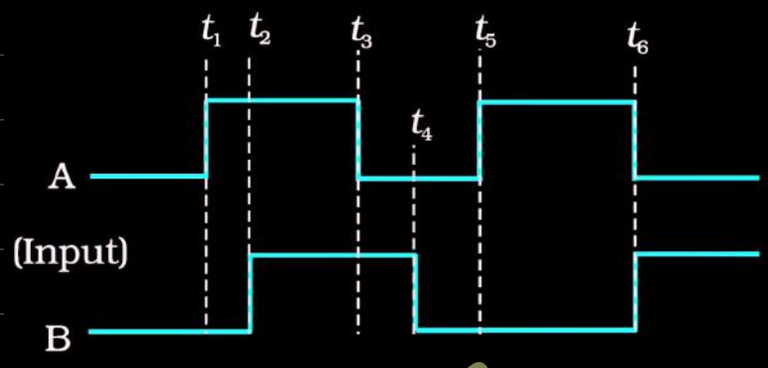
See the switching circuit shown in the figure. When both switches A and B are closed, the bulb becomes short circuited and does not glow (0). When any of the switches is open, current will pass

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through the bulb and it will glow. Thus the circuit follows the truth table of AND gate.

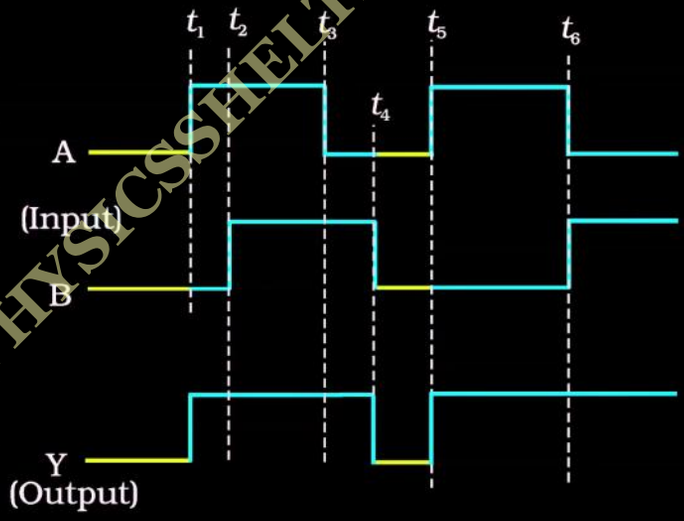
EXAMPLE:

In the adjacent figure A and B represent time varying digital inputs to a OR Gate. Sketch the corresponding digital output of the gate.



Solution:

The desired output is shown in the adjacent figure. We identify the regions where A and B are simultaneously low and mark it as low output. Everywhere else, we mark a high output.

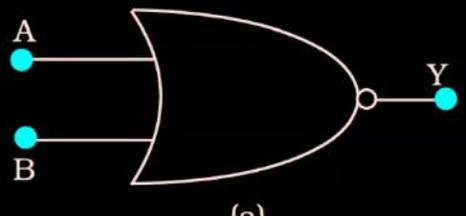


→ Think how would you proceed if it were an AND gate or a NAND gate.

Solution: For these cases we will first identify the regions where both are simultaneously high and mark the corresponding output regions, and then the remaining segments will have the exactly opposite boolean values.

NOR GATE:

$Y = \overline{A+B}$



(a)

Input		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(b)

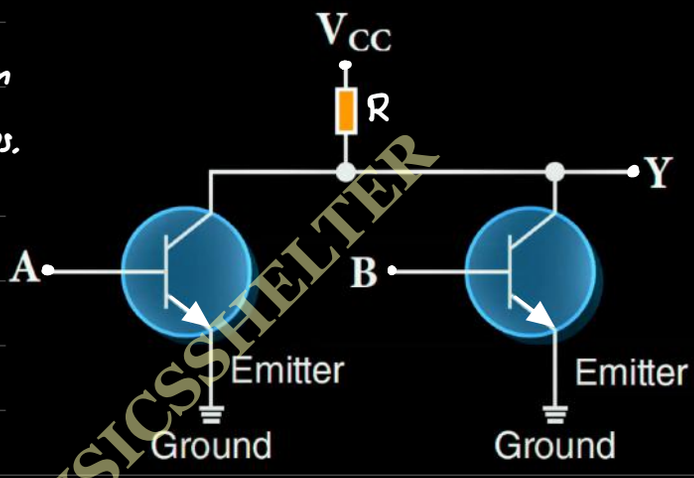
(a) Logic symbol, (b) Truth table of NOR gate.

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A NOR gate is simply an OR gate followed by a not gate accordingly, its output is simply a complement of output of an OR gate. The logic symbol and truth table of a NOR gate are shown in the above figure.

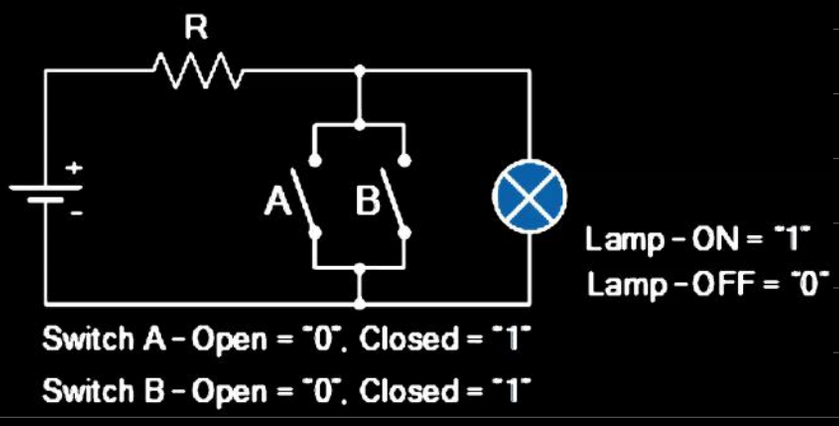
REALIZING NOR GATE USING TRANSISTORS:

The adjacent figure shows realization of a NOR gate using only transistors. Here if any of the inputs A or B are high then that transistor will saturate thus making $V_{CE} = 0$. Thus output will become zero if any of the inputs is 1. When both inputs are low, there is no current through resistor and thus output is same as V_{CC} i.e High. Thus this circuit follows the truth table of NOR gate.



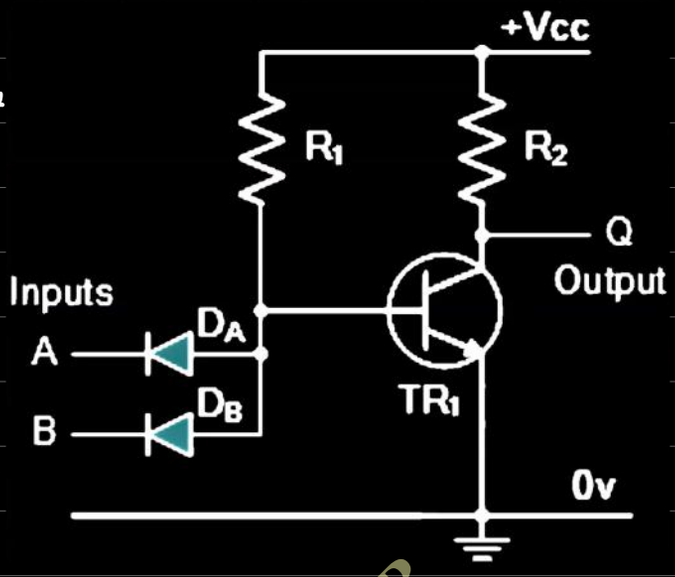
REALIZING A NOR GATE USING SWITCHING CIRCUIT:

The adjacent figure shows a switching circuit to realize a NOR gate. Here if any of the switches is closed the bulb will be shorted and thus not glow (0 output), when both switches are open, the bulb glows due to current from the battery.



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EXAMPLE: Identify the gate represented by the shown circuit in the adjacent figure.

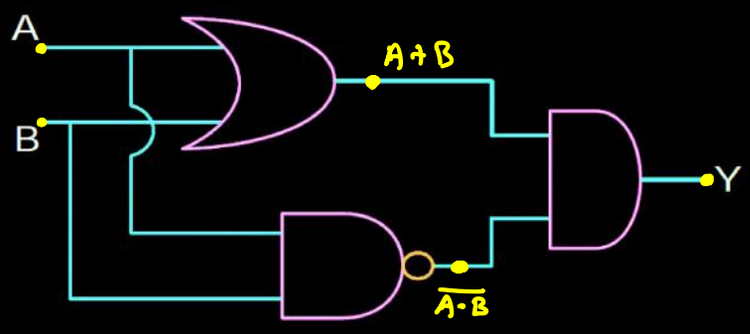


Solution: When any of the inputs A or B is low, the base of transistor gets a low voltage (diode becomes short). Thus transistor becomes cut off and no current flows through R_2. Accordingly, output is high (same as V_{cc}). When both inputs are high, diodes become open circuits and base gets a high voltage due to a small current flowing through R_1 (I_B). Thus transistor gets saturated and $V_{CE} = 0$. Accordingly output becomes zero. Thus this circuit acts like a NAND gate.

BOOLEAN ALGEBRA: The various operations in Boolean algebra are interpreted as follows:

$A \cdot B \equiv$ A AND B, $A + B \equiv$ A OR B and $\bar{A} \equiv$ NOT A.

We can create complex networks of logic gates and then we can express the output in terms of Boolean expression relating output to input. For instance the circuit shown in adjacent figure is represented by the boolean equation:



$Y = (A+B) \cdot (\overline{A \cdot B})$. There are certain theorems in Boolean Algebra that can be used to simplify the boolean expressions which in turn can be used to achieve the same output using a simpler logic gates circuit.

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Boolean Rules For OR, NOT and 'AND' operators:

The rules for operating on various input combinations are listed in adjacent table. These rules follow directly from the truth tables of respective gates.

OR	AND	NOT
0+0=0	0.0=0	$\overline{0} = 1$
0+1=1	0.1=0	$\overline{1} = 0$
1+0=1	1.0=0	
1+1=1	1.1=1	

Laws In Boolean Algebra:

The various laws governing operations in Boolean algebra are listed in adjacent table. These laws also follow directly from the truth tables of respective logic gates. For instance in an OR gate if any input is 1, output is 1. Thus A+1=1. Similarly you can verify all laws.

Commulative laws: A+B=B+A; A.B=B.A

Associative laws:

A+(B+C) = (A+B) +C ; A.(B.C) = (A.B).C

Distributive laws: A.(B+C) = A.B +A.C

Basic OR and AND Relations:

OR	AND
A+0=A	A.0=0
A+1=1	A.1=A
A+A=A	A.A=A
$A+\overline{A}=1$	$A.\overline{A}=0$

Double complement function: $\overline{\overline{A}} = A$

De Morgan's Theorems:

Theorem 1: The complement of the sum of two (or more) variables is equal to the product of complements of the variables i.e. $\overline{A+B} = \overline{A} \bullet \overline{B}$

$$\Rightarrow \overline{A+B} = \overline{A}.\overline{B}$$

Theorem 2 : The complement of the product of two (or more) variables is equal to the sum of complements of the variables i.e $\overline{A \bullet B} = \overline{A} + \overline{B}$.

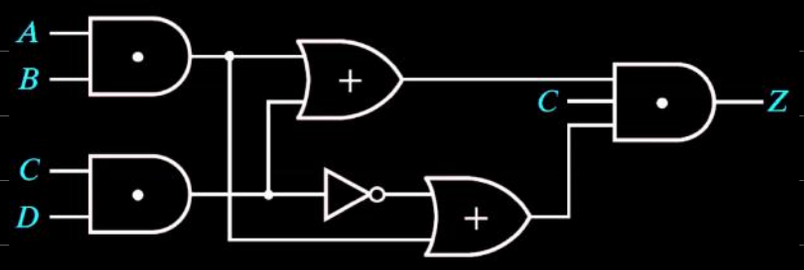
$$\Rightarrow \overline{A.B} = \overline{A} + \overline{B}$$

There are several theorems in Boolean Algebra out of which De Morgan's theorems are amongst most useful theorems for simplifying Boolean expressions. These theorems can be verified either using truth tables or Venn Diagrams (Union is like OR and Intersection is AND).

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Example: Simplify the shown circuit to a single logic gate.



Solution:

$$Z = [(A\cdot B)+(C\cdot D)]\cdot[\overline{C\cdot D}+A\cdot B]\cdot C$$

$$= [(A\cdot B)\cdot\overline{C\cdot D}+(A\cdot B)+0+A\cdot B\cdot C\cdot D]\cdot C \quad (\because \text{Distributive law}).$$

$= (A\cdot B)[\overline{CD}+1+CD]\cdot C = (A\cdot B)(1)C = ABC$. Thus the gate represents simply an AND gate with 3 inputs A, B and C.

Simplification Theorems:

Uniting:	$XY + XY' = X$	$(X + Y)(X + Y') = X$
Absorption:	$X + XY = X$	$X(X + Y) = X$
Elimination:	$X + X'Y = X + Y$	$X(X' + Y) = XY$
Consensus:	$XY + X'Z + YZ = XY + X'Z$	$(X + Y)(X' + Z)(Y + Z) = (X + Y)(X' + Z)$

Proof Of Elimination Theorem:

$X + \bar{X}Y = X + XY + \bar{X}Y = X + Y[X+\bar{X}] = X+Y$ QED.

Proof Of Consensus Theorem:

$XY + \bar{X}Z + YZ = XY + \bar{X}Z + (X+\bar{X})YZ = XY + XYZ + \bar{X}YZ + \bar{X}Z$

$= XY(1+Z) + \bar{X}Z(1+Y) = XY + \bar{X}Z$ QED.

Duality Theorem: If a Boolean equation is true then its dual is also true. A dual is obtained by replacing '.' with '+' and '+' with '.' A 0 is replaced by 1 and 1 is replaced by zero and variables and complements are kept as it is. For instance De Morgan's theorems are dual of each other. $\overline{A\cdot B} = \bar{A}+\bar{B} \iff \overline{A+B} = \bar{A}\cdot\bar{B}$

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EXAMPLE: Simplify the following Boolean Expressions:

(a) $\overline{(A\cdot B)\cdot(\bar{B}+C)}$ (b) $BC+\bar{B}+\bar{B}(A+B)$ (c) $A\left[B+C\left(\overline{AB+AC}\right)\right]$

Solution: (a) $\overline{(A\cdot B)\cdot(\bar{B}+C)} = \overline{A\cdot B\cdot\bar{B}+A\cdot B\cdot C} = \overline{ABC}$ $(\because B\cdot\bar{B}=0)$

5 gates → 2 gates

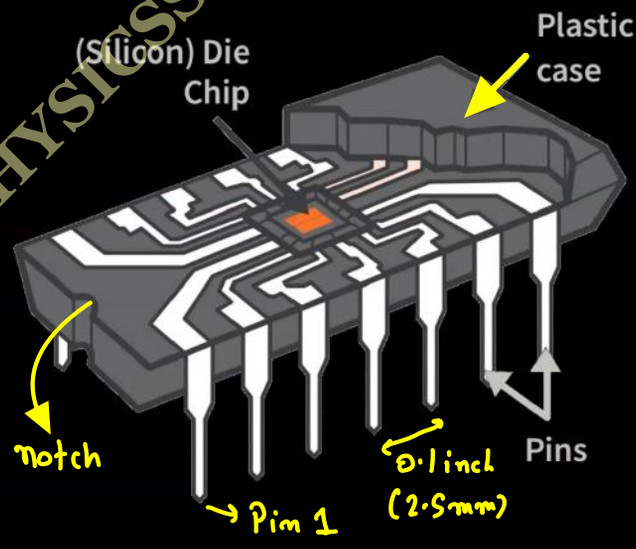
(b) $BC+\bar{B}+\bar{B}(A+B) = BC+\bar{B}[1+A+B] = BC+\bar{B} = \bar{B}+\bar{\bar{B}}C$

$= \bar{B}+C$ (Elimination)

(c) $A\left[B+C\left(\overline{AB+AC}\right)\right] = AB+AC\left[\overline{AB}\cdot\overline{AC}\right] = AB$ $(\because AC\cdot\overline{AC}=0)$.

INTEGRATED CHIP:

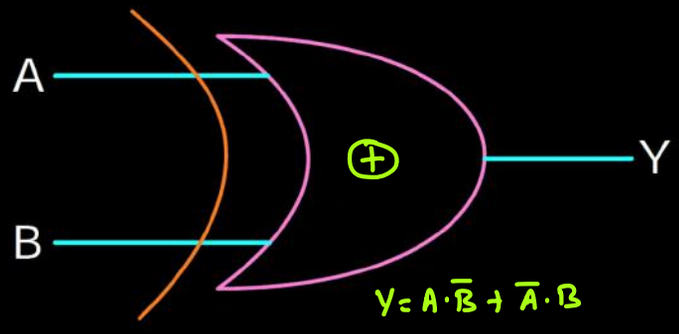
An integrated chip contains an entire circuit consisting of several circuit elements in to a very small chip which could be as small as 1mm x 1mm or even smaller. ICs can be grouped in to two categories (a) linear (or analogue) ICs and (b) digital ICs. Linear ICs process analog signals which change smoothly over a particular range. The output varies almost linearly with input. Digital ICs process the signals which are binary. Depending on integration (number of logic gates), ICs can be classified as Small scale Integration (SSI, $\leq$ 10 logic gates), MSI ($\leq$ 100 logic gates) (medium scale integration), Large Scale Integration ($\leq$ 1000 gates) and VLSI (> 1000 logic gate).

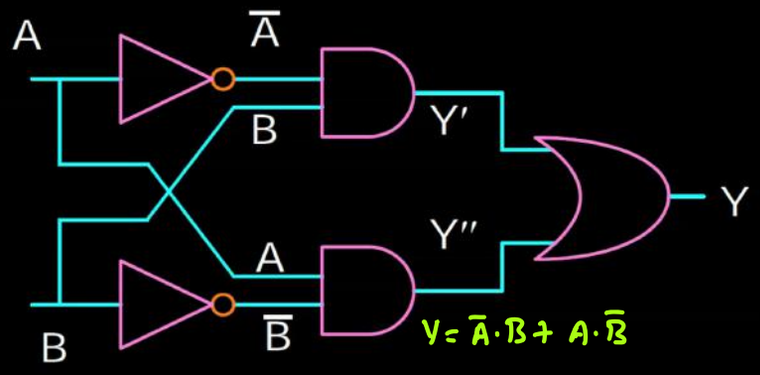


→ IC pin number counting is started anti-clockwise from the notch (see figure).

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XOR GATE: EXCLUSIVE OR ⊕ :





An XOR gate or an exclusive OR gate gives an output value of 1 only if exactly one of its input is 1. In all other cases it gives a zero output. The truth table for an XOR gate is shown in adjacent figure. It can be realized using AND, OR and NOT gates as shown in the figure on top left. The boolean symbol for the Gate is ⊕ thus:

$A \oplus B = A \cdot \bar{B} + \bar{A} \cdot B$.

truth table

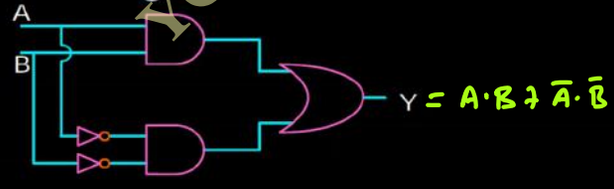
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

XNOR GATE: XNOR gate is obtained by using OR, AND and NOT gates.

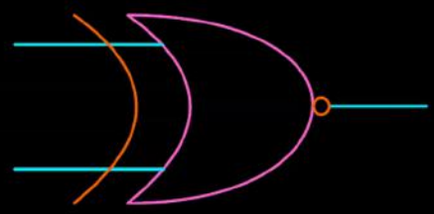
- It is also called exlusive NOR gate.
- The output of a two input XNOR gate is 1 only when both the inputs are same.
- The Boolean equation is $Y = A.B + \bar{A}.\bar{B}$

XNOR gate is inverse of XOR gate.

Diagram



circuit symbol



truth table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

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